Orientalmotor

HM-60016-2

αSTEP

ARL Series

MECHATROLINK-Ⅱ

USER MANUAL

CE

Thank you for purchasing an Oriental Motor product.
This manual describes product handling procedures and safety precautions.

- Please read it thoroughly to ensure safe operation.
- Always keep the manual where it is readily available.

Table of contents

1 Introduction

■ Before use

Only qualified personnel should work with the product.
Use the product correctly after thoroughly reading the section "2 Safety precautions" on p.5.
The product described in this manual has been designed and manufactured for use in general industrial machinery, and must not be used for any other purpose. Oriental Motor Co., Ltd. is not responsible for any damage caused through failure to observe this warning.

■ Structure of the manual

The **ARL** series MECHATROLINK-Ⅱ comes with the manuals specified below.

- **ARL** Series Motor OPERATING MANUAL

  This manual explains the motor functions as well as how to install the motor, among others.

- **ARL** Series MECHATROLINK-Ⅱ Driver OPERATING MANUAL

  This manual explains the driver functions as well as how to install the driver, among others.

- **ARL** Series MECHATROLINK-Ⅱ USER MANUAL (this document)

  This manual explains the function, installation, connection and troubleshooting of the motor and driver.

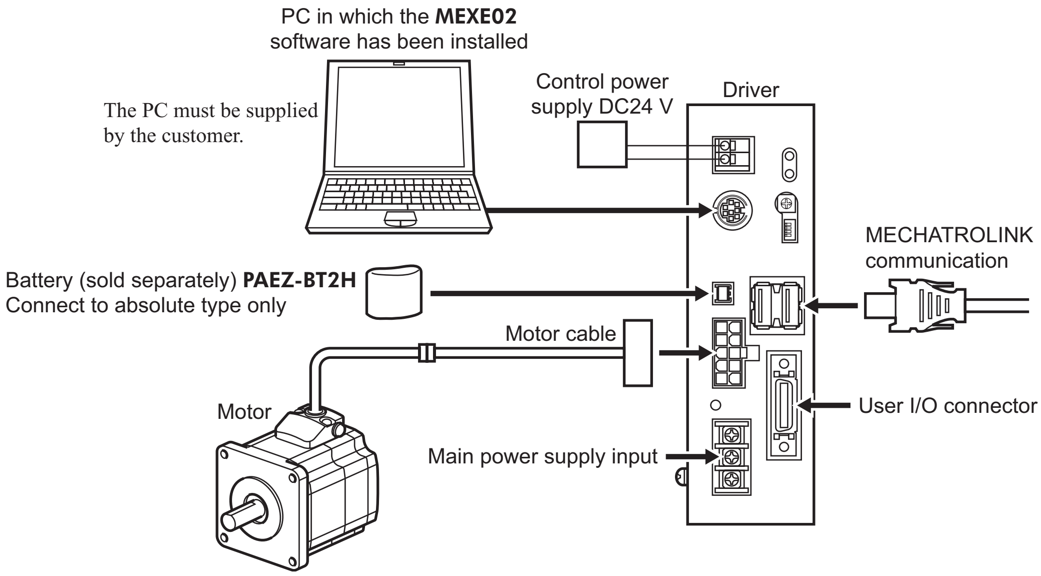
After reading the above manuals, keep them in a convenient place so that you can reference them at any time.

■ Overview of the product

This product is a MECHATROLINK-II compatible unit that can be combined with any **ARL**-series motor.
Set the operating data and parameters using the MECHATROLINK-Ⅱ communication. You can also set parameters using the **MEXE02** data setting software.

■ System configuration

The power supply and MECHATROLINK communication cable are not included and must be provided by the customer.



- If you are using a motor with an electromagnetic brake, be sure to use the supplied cable or an optional extension cable for electromagnetic brake (sold separately). The user must provide a power supply of 24 VDC, exclusively for use with the electromagnetic brake.
- There are two input-power supplies available: single-phase 100-115 V and single-phase 200-230 V.
- To perform return-to-home operation, dedicated sensors will be required.

■ CE Marking

This product is certified under the CE marking standard (Low Voltage Directive, EMC Directive) according to the applicable EN standards.

• For Low Voltage Directive

The product is intended to be incorporated into machinery, so it should be installed within an enclosure.

- Install the product within the enclosure in order to avoid contact with hands.
- Be sure to maintain a protective ground in case hands should make contact with the product. Securely ground the protective grounding terminals of the motor and driver.

Applicable standards

Motor: EN 60950-1, EN 60034-1, EN 60034-5, EN 60664-1
Driver: EN 50178

Installation conditions (EN Standard)

Driver is to be used as a component within other equipment.
Overvoltage category: II
Pollution degree: 2
Protection against electric shock: Class I

• EMC Directive

EMC of this product has been measured according to the configuration illustrated in “Example of motor and driver installation and wiring” on p.21. Be sure to conduct EMC measures with the product assembled in your equipment by referring to “5.7 Installing and wiring in compliance with EMC Directive” on p.19.

■ Hazardous substances

The products do not contain the substances exceeding the restriction values of RoHS Directive (2011/65/EU).

2 Safety precautions

The precautions described below are intended to prevent danger or injury to the user and other personnel through safe, correct use of the product. Use the product only after carefully reading and fully understanding these instructions.

⚠ Warning	Handling the product without observing the instructions that accompany a "Warning" symbol may result in serious injury or death.
⚠ Caution	Handling the product without observing the instructions that accompany a "Caution" symbol may result in injury or property damage.
Note	The items under this heading contain important handling instructions that the user should observe to ensure safe use of the product.

⚠ Warning

General

- Do not use the product in explosive or corrosive environments, in the presence of flammable gases, locations subjected to splashing water, or near combustibles. Doing so may result in fire, electric shock or injury.
- Assign qualified personnel the task of installing, wiring, operating/controlling, inspecting and troubleshooting the product. Failure to do so may result in fire, electric shock, injury or damage to equipment.
- Do not transport, install the product, perform connections or inspections when the power is on. Always turn the power off before carrying out these operations. Failure to do so may result in electric shock.
- The terminals on the driver's front panel marked with ⚠ ⚠ symbol indicate the presence of high voltage. Do not touch these terminals while the power is on to avoid the risk of fire or electric shock.
- When the driver's protection function is triggered, first remove the cause of the alarm and then clear the protection function. Continuing the operation without removing the cause of the problem may cause malfunction of the motor and driver, leading to injury or damage to equipment.

Installation

- To prevent the risk of electric shock, use the motor and driver for class I equipment only.
- Install the motor and driver in their enclosures in order to prevent electric shock or injury.
- Install the motor and driver so as to avoid contact with hands, or ground them to prevent the risk of electric shock.

Connection

- Keep the driver's input power voltage within the specified range to avoid fire and electric shock.
- Connect the cables securely according to the wiring diagram in order to prevent fire and electric shock.
- Do not forcibly bend, pull or pinch the cable. Doing so may fire and electric shock. This will also cause stress to the connecting section and may result in damage to equipment.
- To prevent electric shock, be sure to install the terminal cover (supplied) over the driver's power supply terminals after making connections.

Operation

- Turn off the driver power in the event of a power failure, or the motor may suddenly start when the power is restored and may cause injury or damage to equipment.
- Do not use the servo off command to cut off the electric current while the system is operating. The motor will stop and holding force will be lost, resulting in injury or damage to equipment.
- After entering the battery backup mode due to power outage, the motor does not automatically return to the original position even when the power is restored. Before performing incremental positioning operation, correct the stopped position or perform return-to-home operation.
- If the MECHATROLINK system has experienced a communication error or other problem, refer to the operating manual for the master station to check the status of the MECHATROLINK system and the driver, or contact the Oriental Motor branch or sales office near you. As a precautionary measure, also configure an interlock circuit using a sequence program by allowing the circuit to actuate reflecting the communication status, so that if a system communication error occurs, the entire system including the driver will operate on the safe side.

Maintenance and inspection

- Do not touch the driver immediately after the power is turned off (until the CHARGE LED turns off). The residual voltage may cause electric shock.

Repair, disassembly and modification

- Do not disassemble or modify the motor or driver. This may cause electric shock or injury. Refer all such internal inspections and repairs to the branch or sales office from which you purchased the product.

⚠ Caution

General

- Do not use the motor and driver beyond their specifications, or electric shock, injury or damage to equipment may result.
- Keep your fingers and objects out of the openings in the motor and driver, or fire, electric shock or injury may result.
- Do not touch the motor or driver during operation or immediately after stopping. The surfaces are hot and may cause a skin burn(s).
- To prevent injury or damage to equipment, do not use a battery other than the dedicated optional battery (sold separately).

Installation

- Keep the area around the driver free of combustible materials in order to prevent fire or a skin burn(s).
  To prevent the risk of damage to equipment, leave nothing around the driver that would obstruct ventilation.

Operation

- Use a motor and driver only in the specified combination. An incorrect combination may cause a fire.
- Provide an emergency stop device or emergency-stop circuit external to the equipment so that the entire equipment will operate safely in the event of a system failure or malfunction. Failure to do so may result in injury.
- Before supplying power to the driver, turn all control inputs to the driver to OFF. Otherwise when the power is turned on, the motor may start suddenly and cause injury or damage to equipment.
- To prevent bodily injury, do not touch the rotating parts (output shaft) of the motor during operation.
- If the motor’s output shaft must be moved directly by hand, do so after turning off the driver’s main power supply or the motor current. Failure to do so may result in injury.
- For the control power supply, use a power supply with reinforced insulation provided on the primary side, and provide it separately from the power supply for the electromagnetic brake. Failure to do so may result in electric shock.
- Immediately when trouble has occurred, stop running the motor and turn off the driver power. Failure to do so may result in fire, electric shock or injury.
- Static electricity may cause the driver to malfunction or suffer damage. While the driver is receiving power, do not touch the driver. Use only an insulated screwdriver to adjust the driver's switches.

Inspection

- To prevent the risk of electric shock, do not touch the terminals while measuring the insulation resistance or conducting a voltage-resistance test.

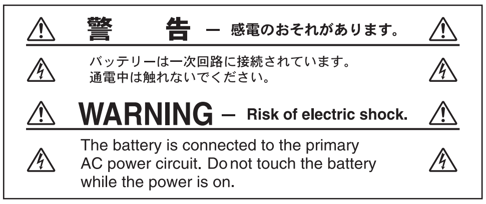
Disposal

- To dispose of the motor or driver, disassemble it into parts and components as much as possible and dispose of individual parts/components as industrial waste. The product uses nickel-cadmium batteries. Dispose of the used batteries in accordance with local laws and regulations. If you have any questions regarding disposal of the product, please contact any Oriental Motor branch or sales office.

■ Warning display of the battery

Words of warning on handling are displayed on the battery. Be sure to observe the information written on the label/plate when handling the battery.

- Connecting warning label for electric shock (battery)



Material: Polypropylene

- Battery rating plate

Material: PET

■ Handling the battery

Always observe the following items when using the battery. Failure to handle the battery correctly may cause electric shock, and the battery to leak or burst, resulting in injury or equipment damage.

⚠ Warning

- The battery is connected to the primary circuit, so do not touch the battery while the power is on. Do not touch these terminals while the power is on.
- Do not heat the battery or throw it into a fire.
- Never short-circuit the battery or connect the positive and negative terminals in reverse.
- When carrying/storing the battery, do not place it together with metal necklaces, hairpins, coins, keys or other conductive objects.
- When storing the battery, store it away from direct sunlight in a place not subject to high temperature or high humidity.
- Do not disassemble or modify the battery.
- Do not apply solder directly to the battery.
- Do not cut or modify the battery cable.
- Use a dedicated driver to charge the battery.
- The battery has a vent structure for the release of internal gas. Do not apply a strong force to the battery, since it may cause this structure to deform.
- When installing the battery into the machine, never place it inside a sealed structure. The battery sometimes generates gas, which, if trapped, may cause a burst or an explosion due to ignition.
- The battery contains an alkali solution. If the alkali solution comes in contact with the skin or clothes, flush the area thoroughly with clean water. If the alkali solution gets into the eyes, do not rub. Flush the eyes thoroughly with clean water and seek immediate medical attention.
- Do not use the battery if there is leakage, discoloration, deformation or another abnormality.
- Do not immerse the battery in water or seawater, nor allow it to become wet. Doing so may cause the battery to generate heat or rust.
- Do not scratch the battery. A scratched battery easily causes shorting, resulting in leakage, heat generation or bursting.
- Disconnect the battery connector if the controller is not turned on for an extended period exceeding the data retention period. Failure to do so may cause the battery fluid to leak or battery performance to drop.
- When storing the battery, disconnect the battery connector. Store the battery in a place with an ambient temperature of −20 to +45 °C (−4 to +113 °F) [or −20 to +35 °C (−4 to +95 °F) if the battery is to be stored for 3 months or more] where humidity is not excessively high. Failure to heed this instruction may result in leaked battery fluid, rusting or performance deterioration of the battery.

3 Precautions for use

This section covers limitations and requirements the user should consider when using the product.

- **Use the supplied cable to connect the motor and driver.**

  Always use the supplied cable to connect the motor and driver.
  If a flexible cable or cable of 3 m (9.8 ft.) or longer is to be used, the appropriate cable must be purchased separately. See p.73 for details.

- **Conduct the insulation resistance measurement or withstand voltage test separately on the motor and the driver.**

  Conducting the insulation resistance measurement or withstand voltage test with the motor and driver connected may result in injury or damage to equipment.

- **Do not apply an radial load and axial load in excess of the specified permissible limit.**

  Be sure to operate the motor within the specified permissible limit of radial load and axial load. Operating it under an excessive radial load and axial load may damage the motor bearings (ball bearings). See p.15 for details.

- **Operate the motor with a surface temperature not exceeding 100 °C (212 °F).**

  The driver has an overheat-protection function, but the motor has no such feature. The motor case's surface temperature may exceed 100 °C (212 °F) under certain conditions (ambient temperature, operating speed, duty cycle, etc.). Keeping the surface temperature of the motor casing below 100 °C (212 °F) will also maximize the life of the motor bearings (ball bearings).
  Use the geared type motor in a condition where the gear case temperature does not exceed 70 °C (158 °F), in order to prevent deterioration of grease in the gear.

- **Maximum static torque at excitation**

  Maximum static torque at excitation represents a value obtained when the motor is excited using a rated current. When combined with a dedicated driver and while the motor is stopped motor-temperature increases are suppressed due to a current-reduction of approximately 50% by the current-cutback function. Acceleration and operation at the maximum static torque at excitation is possible in start-up, but it has approximately 50% holding power after it has stopped. When selecting a motor for your application, consider the fact that the holding power will be reduced to approximately 50% after the motor has stopped.

- **Use an electromagnetic brake type for a vertical applications.**

  - When the motor is used in a vertical application, use an electromagnetic brake type to hold the load in position. To hold the load in position, apply the electromagnetic brake only after the motor has stopped.
  - Do not use the brake to bring the moving motor to a halt. Repeated braking for such a purpose will wear the brake hub excessively, causing its holding ability to drop.
  - Since the electromagnetic brake is of the non-excitation type, it can also be used to hold the load in position upon the occurrence of a power failure. However, this is not a secure means of holding the load. Do not use the electromagnetic brake as a safety brake.
  - When the driver's protective function is triggered, the motor stops as the current is turned off. The user must set a controller sequence that will cut off the power to the electromagnetic brake and hold the load in position upon detecting an "OFF" ALARM output.

- **Connecting an electromagnetic brake motor**

  When using the electromagnetic brake motor, provide separate power supplies for the I/O signals and the electromagnetic brake.

- **Install the driver in a vertical orientation.**

  The driver's heat-dissipation function is designed according to vertical orientation. Installing the driver in any other orientation may shorten the life of electronic parts due to temperature increases within the driver.

- **Preventing electrical noise**

  See "5.7 Installing and wiring in compliance with EMC Directive" on p.19 for measures with regard to noise.

- **Preventing leakage current**

  Stray capacitance exists between the driver's current-carrying line and other current-carrying lines, the earth and the motor, respectively. A high-frequency current may leak out through such capacitance, having a detrimental effect on the surrounding equipment. The actual leakage current value depends on the driver's switching frequency, the length of wiring between the driver and motor, and so on.
  When installing a leakage current breaker, use the following products, for instance, which have high-frequency signal protection:
  Mitsubishi Electric Corporation: NV series
  Fuji Electric FA Components & Systems Co., Ltd.: EG and SG series

- **Maximum torque of geared type motor**

  Always operate the geared type motor under a load not exceeding the maximum torque. If the load exceeds the maximum torque, the gear will be damaged.

- **About grease of geared motor**

  On rare occasions, a small amount of grease may ooze out from the geared motor. If there is concern over possible environmental damage resulting from the leakage of grease, check for grease stains during regular inspections. Alternatively, install an oil pan or other device to prevent leakage from causing further damage. Oil leakage may lead to problems in the customer's equipment or products.

- **Do not perform push-motion operation with geared type motors.**

  The motor or gearhead may be damaged as a result.

- **Storing data in the NV memory**
  - Do not turn off the control power while data is being written to the NV memory and for at least 5 seconds after writing is completed. Doing so may generate a memory error alarm.
  - Rewrite life of the NV memory is approx. 100,000 writes. Note that if the same value is written to the same parameter, the parameter will not be overwritten and thus the write counter will not be incremented.

- **Battery backup (absolute type)**

  Always charge the battery before using by referring to p.24. Data can be backed up with a charged battery for a period of up to 15 days.
  Data may not be backed up correctly in the following conditions:
  - The multi-rotation operation range (−41943 to +41943 revolutions) is exceeded.
    The multi-rotation means repeating continuous operations or positioning operations in the same direction. One revolution in multi-rotation operation refers to one revolution of the motor shaft.
  - The electronic gear is changed after a movement by positioning operation, etc.
  - The internal counter is reset to zero by turning ON the PRESET input during operation.

  Note
  - The range within which multi-rotation operation is permitted is −41943 to +41943 revolutions, where 0 represents the electrical home position.
  - If the multi-rotation operation range (−41943 to +41943 revolutions) is exceeded, cycling the control power will generate an absolute position loss error. The absolute position loss alarm can be reset by changing the value of the absolute position loss error reset parameter from 0 to 1. After the alarm has been reset, perform a return-to-home operation or use a coordinate-system setting command to confirm the coordinates.

- **Note on connecting a power supply whose positive terminal is grounded**

  When grounding the positive terminal of the power supply, do not connect any equipment (PC, etc.) whose negative terminal is grounded. Doing so may cause the driver and these equipment to short, damaging both. Use MECHATROLINK-II communication to set data, etc.

4 Preparation

This section covers the points to be checked along with the names and functions of respective parts.

4.1 Checking the product

Verify that the items listed below are included. Report any missing or damaged items to the branch or sales office from which you purchased the product.
Verify the model number of the purchased unit against the number shown on the package label.
Check the model number of the motor and driver against the number shown on the nameplate.
The unit models and corresponding motor/driver combinations are listed on "4.2 Combinations of motors and drivers".

- Motor .... 1 unit
- Driver.. 1 unit
- Driver mounting brackets .... 2 pcs.
- Screws for driver mounting brackets (M3) ... 4 pcs.
- CN5 connector (26 pins)............................... 1 pc.
- Cable for motor... 1 pc.
- Cable for electromagnetic brake .... 1 pc. (supplied with electromagnetic brake motor)
- Varistor.. 1 pc. (supplied with electromagnetic brake motor)
- OPERATING MANUAL Motor .... 1 copy
- OPERATING MANUAL Driver................ 1 copy
- USER MANUAL and Teaching Pendant SETTING MANUAL (CD-ROM)................ 1 pc.
- Parallel key .... 1 pc. (supplied with geared types; except for the **ARL46TH** and **ARL66TH**)

4.2 Combinations of motors and drivers

- The box (☐) will be filled with **A** (single shaft), **B** (double shaft) or **M** (with electromagnetic brake.)
  In **ARL911☐AM-○** and **ARL911☐CM-○**, ☐ indicates **A** (single shaft) or **B** (double shaft).
  ☐ indicates **A** (single shaft) or **M** (with electromagnetic brake) for geared type.
- ○ indicates the length of the supplied cable (**1**, **2**, **3**).

• Standard type

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL46☐AM-○	ARLM46☐A	ARLD13A-AM2
ARL66☐AM-○	ARLM66☐A	ARLD24A-AM2
ARL69☐AM-○	ARLM69☐A	ARLD30D-AM2
ARL98☐AM-○	ARLM98☐A	ARLD30A-AM2
ARL911☐AM-○	ARLM911☐A	ARLD30E-AM2

• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL46☐CM-○	ARLM46☐C	ARLD07A-CM2
ARL66☐CM-○	ARLM66☐C	ARLD12A-CM2
ARL69☐CM-○	ARLM69☐C	ARLD16D-CM2
ARL98☐CM-○	ARLM98☐C	ARLD16A-CM2
ARL911☐CM-○	ARLM911☐C	ARLD20A-CM2

• TH geared type

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL46☐AM-T3.6-○	ARLM46☐A-T3.6	ARLD13B-AM2
ARL46☐AM-T7.2-○	ARLM46☐A-T7.2	
ARL46☐AM-T10-○	ARLM46☐A-T10	
ARL46☐AM-T20-○	ARLM46☐A-T20	ARLD13C-AM2
ARL46☐AM-T30-○	ARLM46☐A-T30	
ARL66☐AM-T3.6-○	ARLM66☐A-T3.6	ARLD24B-AM2
ARL66☐AM-T7.2-○	ARLM66☐A-T7.2	
ARL66☐AM-T10-○	ARLM66☐A-T10	
ARL66☐AM-T20-○	ARLM66☐A-T20	ARLD24C-AM2
ARL66☐AM-T30-○	ARLM66☐A-T30	
ARL98☐AM-T3.6-○	ARLM98☐A-T3.6	ARLD30A-AM2
ARL98☐AM-T7.2-○	ARLM98☐A-T7.2	
ARL98☐AM-T10-○	ARLM98☐A-T10	

• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL46☐CM-T3.6-○	ARLM46☐C-T3.6	ARLD07B-CM2
ARL46☐CM-T7.2-○	ARLM46☐C-T7.2	
ARL46☐CM-T10-○	ARLM46☐C-T10	
ARL46☐CM-T20-○	ARLM46☐C-T20	ARLD07C-CM2
ARL46☐CM-T30-○	ARLM46☐C-T30	
ARL66☐CM-T3.6-○	ARLM66☐C-T3.6	ARLD12B-CM2
ARL66☐CM-T7.2-○	ARLM66☐C-T7.2	
ARL66☐CM-T10-○	ARLM66☐C-T10	
ARL66☐CM-T20-○	ARLM66☐C-T20	ARLD12C-CM2
ARL66☐CM-T30-○	ARLM66☐C-T30	
ARL98☐CM-T3.6-○	ARLM98☐C-T3.6	ARLD16A-CM2
ARL98☐CM-T7.2-○	ARLM98☐C-T7.2	
ARL98☐CM-T10-○	ARLM98☐C-T10	

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL98□AM-T20-○	ARLM98□A-T20	ARLD30C-AM2
ARL98□AM-T30-○	ARLM98□A-T30	

• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL98□CM-T20-○	ARLM98□C-T20	ARLD16C-CM2
ARL98□CM-T30-○	ARLM98□C-T30	

- **PS** geared type

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL46□AM-PS5-○	ARLM46□A-PS5	ARLD13A-AM2
ARL46□AM-PS7-○	ARLM46□A-PS7	
ARL46□AM-PS10-○	ARLM46□A-PS10	
ARL46□AM-PS25-○	ARLM46□A-PS25	
ARL46□AM-PS36-○	ARLM46□A-PS36	ARLD13B-AM2
ARL46□AM-PS50-○	ARLM46□A-PS50	ARLD13C-AM2
ARL66□AM-PS5-○	ARLM66□A-PS5	ARLD24A-AM2
ARL66□AM-PS7-○	ARLM66□A-PS7	
ARL66□AM-PS10-○	ARLM66□A-PS10	
ARL66□AM-PS25-○	ARLM66□A-PS25	ARLD24B-AM2
ARL66□AM-PS36-○	ARLM66□A-PS36	ARLD24C-AM2
ARL66□AM-PS50-○	ARLM66□A-PS50	
ARL98□AM-PS5-○	ARLM98□A-PS5	ARLD30A-AM2
ARL98□AM-PS7-○	ARLM98□A-PS7	
ARL98□AM-PS10-○	ARLM98□A-PS10	
ARL98□AM-PS25-○	ARLM98□A-PS25	
ARL98□AM-PS36-○	ARLM98□A-PS36	
ARL98□AM-PS50-○	ARLM98□A-PS50	ARLD30B-AM2

• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL46□CM-PS5-○	ARLM46□C-PS5	ARLD07A-CM2
ARL46□CM-PS7-○	ARLM46□C-PS7	
ARL46□CM-PS10-○	ARLM46□C-PS10	
ARL46□CM-PS25-○	ARLM46□C-PS25	
ARL46□CM-PS36-○	ARLM46□C-PS36	ARLD07B-CM2
ARL46□CM-PS50-○	ARLM46□C-PS50	ARLD07C-CM2
ARL66□CM-PS5-○	ARLM66□C-PS5	ARLD12A-CM2
ARL66□CM-PS7-○	ARLM66□C-PS7	
ARL66□CM-PS10-○	ARLM66□C-PS10	
ARL66□CM-PS25-○	ARLM66□C-PS25	ARLD12B-CM2
ARL66□CM-PS36-○	ARLM66□C-PS36	ARLD12C-CM2
ARL66□CM-PS50-○	ARLM66□C-PS50	
ARL98□CM-PS5-○	ARLM98□C-PS5	ARLD16A-CM2
ARL98□CM-PS7-○	ARLM98□C-PS7	
ARL98□CM-PS10-○	ARLM98□C-PS10	
ARL98□CM-PS25-○	ARLM98□C-PS25	
ARL98□CM-PS36-○	ARLM98□C-PS36	
ARL98□CM-PS50-○	ARLM98□C-PS50	ARLD16B-CM2

- **PN** geared type

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL46□AM-N5-○	ARLM46□A-N5	ARLD13A-AM2
ARL46□AM-N7.2-○	ARLM46□A-N7.2	
ARL46□AM-N10-○	ARLM46□A-N10	
ARL66□AM-N5-○	ARLM66□A-N5	ARLD24A-AM2
ARL66□AM-N7.2-○	ARLM66□A-N7.2	
ARL66□AM-N10-○	ARLM66□A-N10	
ARL66□AM-N25-○	ARLM66□A-N25	ARLD24B-AM2
ARL66□AM-N36-○	ARLM66□A-N36	ARLD24C-AM2
ARL66□AM-N50-○	ARLM66□A-N50	
ARL98□AM-N5-○	ARLM98□A-N5	ARLD30A-AM2
ARL98□AM-N7.2-○	ARLM98□A-N7.2	
ARL98□AM-N10-○	ARLM98□A-N10	
ARL98□AM-N25-○	ARLM98□A-N25	
ARL98□AM-N36-○	ARLM98□A-N36	
ARL98□AM-N50-○	ARLM98□A-N50	ARLD30B-AM2

• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL46□CM-N5-○	ARLM46□C-N5	ARLD07A-CM2
ARL46□CM-N7.2-○	ARLM46□C-N7.2	
ARL46□CM-N10-○	ARLM46□C-N10	
ARL66□CM-N5-○	ARLM66□C-N5	ARLD12A-CM2
ARL66□CM-N7.2-○	ARLM66□C-N7.2	
ARL66□CM-N10-○	ARLM66□C-N10	
ARL66□CM-N25-○	ARLM66□C-N25	ARLD12B-CM2
ARL66□CM-N36-○	ARLM66□C-N36	ARLD12C-CM2
ARL66□CM-N50-○	ARLM66□C-N50	
ARL98□CM-N5-○	ARLM98□C-N5	ARLD16A-CM2
ARL98□CM-N7.2-○	ARLM98□C-N7.2	
ARL98□CM-N10-○	ARLM98□C-N10	
ARL98□CM-N25-○	ARLM98□C-N25	
ARL98□CM-N36-○	ARLM98□C-N36	
ARL98□CM-N50-○	ARLM98□C-N50	ARLD16B-CM2

- Harmonic geared type

• Single-phase 100-115 V

Unit model	Motor model	Driver model
ARL46□AM-H50-○	ARLM46□A-H50	ARLD13A-AM2
ARL46□AM-H100-○	ARLM46□A-H100	
ARL66□AM-H50-○	ARLM66□A-H50	ARLD24B-AM2
ARL66□AM-H100-○	ARLM66□A-H100	ARLD24C-AM2
ARL98□AM-H50-○	ARLM98□A-H50	ARLD30B-AM2
ARL98□AM-H100-○	ARLM98□A-H100	

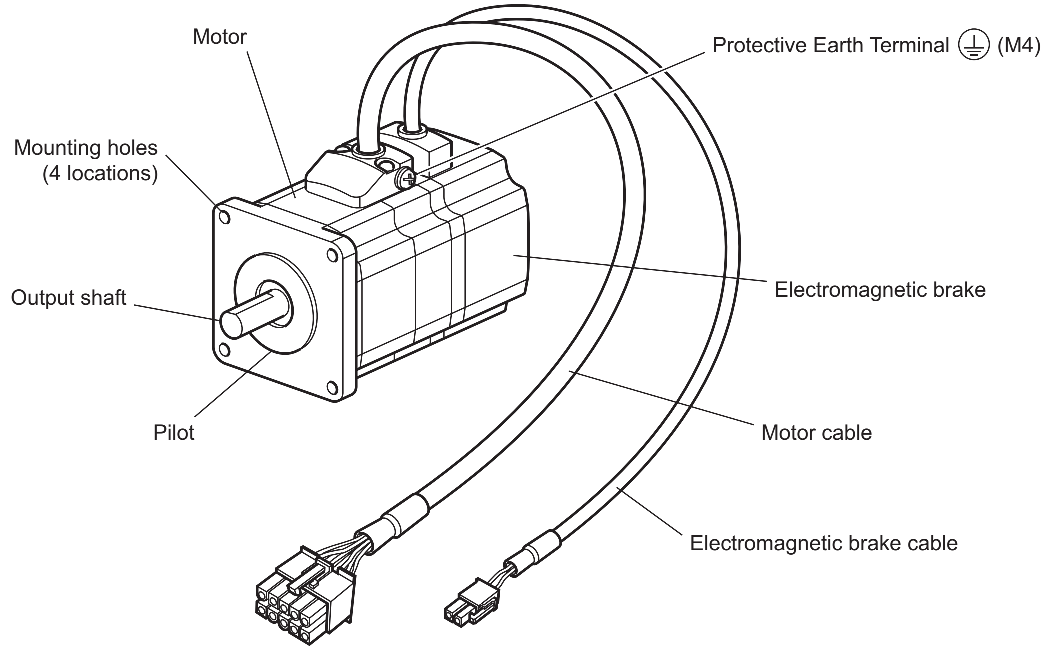
• Single-phase 200-230 V

Unit model	Motor model	Driver model
ARL46□CM-H50-○	ARLM46□C-H50	ARLD07A-CM2
ARL46□CM-H100-○	ARLM46□C-H100	
ARL66□CM-H50-○	ARLM66□C-H50	ARLD12B-CM2
ARL66□CM-H100-○	ARLM66□C-H100	ARLD12C-CM2
ARL98□CM-H50-○	ARLM98□C-H50	ARLD16B-CM2
ARL98□CM-H100-○	ARLM98□C-H100	

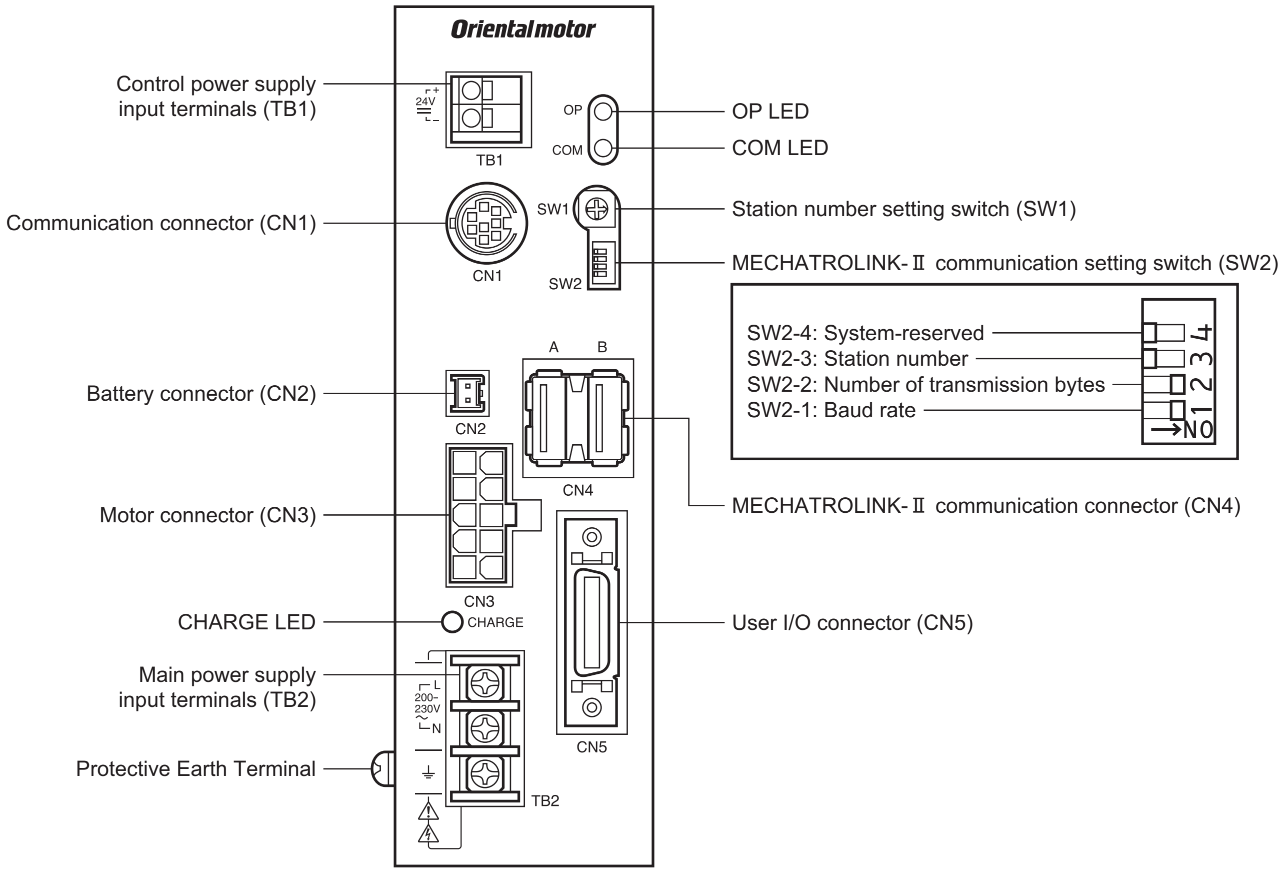
4.3 Names and functions of parts

■ Motor

Illustration shows the standard type with electromagnetic brake.



■ Driver



Name	Description	Ref.
Control power supply input terminals (TB1)	Connects the driver's control circuit power. +: 24 VDC±10% −: GND	p.24
Communication connector (CN1)	Connects PC in which the **MEXE02** software has been installed.	p.27
Battery connector (CN2)	Connects a battery to backup absolute positions.	p.24
Motor connector (CN3)	Connects the motor.	p.22
CHARGE LED (red)	Red when main power is supplied. Once the main power is turned off, the LED will turn off when the residual voltage in the driver drops to a safe level (which takes approximately 6 minutes).	−
Main power supply input terminals (TB2)	Connects the motor driving power supply. Single-phase 100 V type: 100-115 VAC 50/60 Hz Single-phase 200 V type: 200-230 VAC 50/60 Hz	p.23
Protective Earth Terminal ⏚	Make ground connection by installing a grounding wire of AWG18 (0.75 mm^2) or larger.	p.23
OP LED (green/red)	LED shows the condition of the control power supply input and alarm generation. OFF: when control power is not supplied. Green (steady): when operating the driver normally. Red (blinking): when an alarm is generated. Red (steady): When the driver is faulty.	p.67
COM LED (green/red)	LED shows the condition of the MECHATROLINK communication. OFF: when MECHATROLINK communication is not established (other than phase 2, 3). Green lit: when MECHATROLINK communication in progress (phase 2, 3). Red lit: when a communication error is generated.	p.67
Station number setting switch (SW1)	Set the driver's station number in combination with SW2-3. When changing the switch, cycle the control power supply. Factory setting: 41h	p.30
MECHATROLINK-Ⅱ communication setting switch (SW2)	Set the specification of MECHATROLINK communication. When changing the switch, cycle the control power supply.	p.30
MECHATROLINK-Ⅱ communication connector (CN4)	Connects the MECHATROLINK communication cable.	p.27
User I/O connector (CN5)	Connects the User I/O.	p.25

5 Installation

This chapter explains the installation location and installation method of the motor and driver, as well as the method for installing a battery. Also covered in this section are the installation and wiring methods that are in compliance with the relevant EMC Directives.

5.1 Location for installation

The motor and driver are designed and manufactured for installation in equipment. Install them in a well-ventilated location that provides easy access for inspection. The location must also satisfy the following conditions:

- Inside an enclosure that is installed indoors (provide vent holes)
- Operating ambient temperature
  Motor: 0 to +50 °C (+32 to +122 °F) (non-freezing)
  Harmonic geared type: 0 to +40 °C (+32 to +104 °F) (non-freezing)
  Driver: 0 to +40 °C (+32 to +104 °F) (non-freezing)
- Operating ambient humidity 85% or less (non-condensing)
- Area that is free of an explosive nature or toxic gas (such as sulfuric gas) or liquid
- Area not exposed to direct sun
- Area free of excessive amount dust, iron particles or the like
- Area not subject to splashing water (rain, water droplets), oil (oil droplets) or other liquids
- Area free of excessive salt
- Area not subject to continuous vibration or excessive shocks
- Area free of excessive electromagnetic noise (from welders, power machinery, etc.)
- Area free of radioactive materials, magnetic fields or vacuum
- 1000 m (3300 ft.) or lower above sea level

5.2 Installing the motor

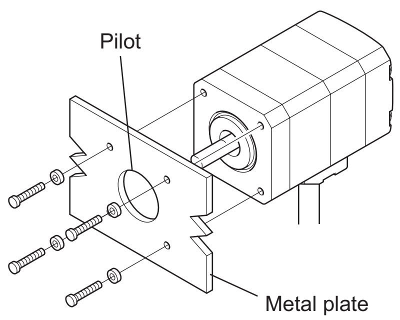
■ Installation direction

The motor can be installed in any direction.

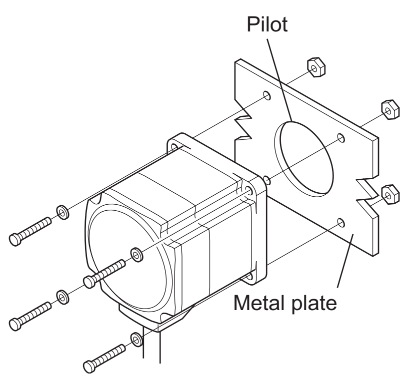
■ Installation method

To allow for heat dissipation and prevent vibration, install the motor on a metal surface of sufficient strength.
When installing the motor, secure it with four bolts (not supplied) through the four mounting holes.
Do not leave a gap between the motor and metal plate.

- Installation method A



- Installation method B



Note Insert the pilot located on the motor's installation surface into the mounting plate's.

Type	Frame size [mm (in.)]	Nominal size	Tightening torque [N·m (oz-in)]	Effective depth of bolt [mm (in.)]	Installation method
Standard	42 (1.65)	M3	1 (142)	4.5 (0.177)	A
	60 (2.36)	M4	2 (280)	−	B
	85 (3.35)	M6	3 (420)	−	
TH geared	42 (1.65) 60 (2.36)	M4	2 (280)	8 (0.315)	A
	90 (3.54)	M8	4 (560)	15 (0.591)	
PS geared **PN** geared Harmonic geared (**ARL46**, **ARL66**)	42 (1.65)	M4	2 (280)	8 (0.315)	
	60 (2.36)	M5	2.5 (350)	10 (0.394)	
	90 (3.54)	M8	4 (560)	15 (0.591)	
Harmonic geared (**ARL98**)	90 (3.54)	M8	4 (560)	−	B

5.3 Installing a load

When connecting a load to the motor, align the centers of the motor's output shaft and load shaft. Also, keep the radial load and axial load to the permissible values or below.
Optional flexible couplings are available (sold separately).

Note
- When coupling the load to the motor, pay attention to the centering of the shafts, belt tension, parallelism of the pulleys, and so on. Securely tighten the coupling and pulley set screws.
- Be careful not to damage the output shaft or bearings (ball bearing) when installing a coupling or pulley to the motor's output shaft.
- Do not modify or machine the motor's output shaft. Doing so may damage the bearings and destroy the motor.
- When inserting a parallel key into the gear output shaft, do not apply excessive force by using a hammer or similar tool. Application of strong impact may damage the output shaft or bearings.

- Using a coupling

Align the centers of the motor's output shaft and load shaft in a straight line.

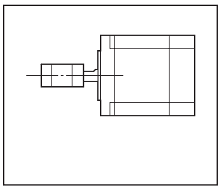
- Using a belt drive

Align the motor's output shaft and load shaft in parallel with each other, and position both pulleys so that the line connecting their centers is at a right angle to the shafts.

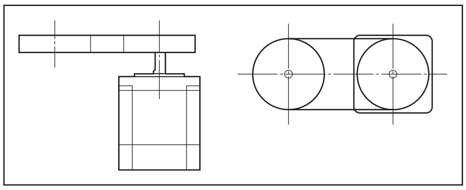
- Using a gear drive

Align the motor's output shaft and gear shaft in parallel with each other, and let the gears mesh at the center of the tooth widths.

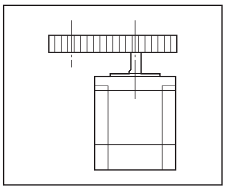
• Using a coupling



• Using a belt drive



• Using a gear drive

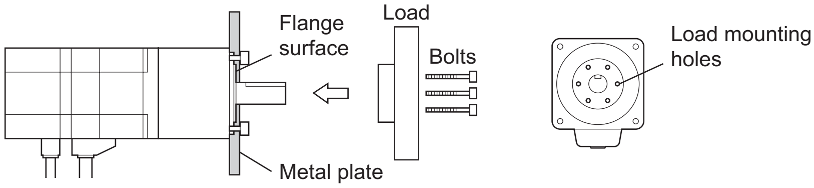


- Using a fastening key (geared motor)

Connect a load to the gear output shaft having a key groove, first provide a key groove on the load and fix the load with the gear output shaft using the supplied key.

- Installing on the flange surface (harmonic geared type)

With a harmonic geared type (except an **ARL98**), a load can be installed directly to the gear using the load mounting holes provided on the flange surface.



Unit model	Nominal size	Number of bolts	Tightening torque [N·m (oz-in)]	Effective depth of bolt [mm (in.)]
ARL46	M3	6	1.4 (198)	5 (0.20)
ARL66	M4	6	2.5 (350)	6 (0.236)

Note
- When installing a load on the flange surface, the load cannot also be affixed using the keyway (or milled surface) in the output shaft.
- Design an appropriate installation layout so that the load will not contact the metal plate or bolts used for installing the motor.

5.4 Permissible radial load and permissible axial load

The radial load and the axial load on the motor's output shaft must be kept under the permissible values listed below.

Note
- Failure due to fatigue may occur when the motor bearings and output shaft are subject to repeated loading by an radial or axial load that is in excess of the permissible limit.
- With the double-shaft type products, the output shaft on the opposite side of the motor output shaft is for installing the slit plate. Do not apply load torque, radial load or axial load on this output shaft.
- The permissible radial load and permissible axial load of the **PS** geared type and **PN** geared type represent the value that the service life of the gear part satisfies 20,000 hours when either of the radial load or axial load is applied to the gear output shaft.

Type	Frame size [mm (in.)]	Unit model	Gear ratio	Permissible radial load [N (lb.)]					Permissible axial load [N (lb.)]
				Distance from the tip of motor output shaft					
				0 mm (0 in.)	5 mm (0.20 in.)	10 mm (0.39 in.)	15 mm (0.59 in.)	20 mm (0.79 in.)	
Standard	42 (1.65)	**ARL46**	−	35 (7.8)	44 (9.9)	58 (13)	85 (19.1)	−	15 (3.3)
	60 (2.36)	**ARL66** **ARL69**		90 (20)	100 (22)	130 (29)	180 (40)	270 (60)	30 (6.7)
	85 (3.35)	**ARL98** **ARL911**		260 (58)	290 (65)	340 (76)	390 (87)	480 (108)	60 (13.5)
TH geared	42 (1.65)	**ARL46**	3.6, 7.2, 10, 20, 30	10 (2.2)	14 (3.1)	20 (4.5)	30 (6.7)	−	15 (3.3)
	60 (2.36)	**ARL66**		70 (15.7)	80 (18)	100 (22)	120 (27)	150 (33)	40 (9)
	90 (3.54)	**ARL98**		220 (49)	250 (56)	300 (67)	350 (78)	400 (90)	100 (22)
PS geared	42 (1.65)	**ARL46**	5	70 (15.7)	80 (18)	95 (21)	120 (27)	−	100 (22)
			7.2	80 (18)	90 (20)	110 (24)	140 (31)	−	
			10	85(19.1)	100 (22)	120 (27)	150 (33)	−	
			25	120 (27)	140 (31)	170 (38)	210 (47)	−	
			36	130(29)	160 (36)	190 (42)	240 (54)	−	
			50	150 (33)	170 (38)	210 (47)	260 (58)	−	
	60 (2.36)	**ARL66**	5	170 (38)	200 (45)	230 (51)	270 (60)	320 (72)	200 (45)
			7.2	200 (45)	220 (49)	260 (58)	310 (69)	370 (83)	
			10	220 (49)	250 (56)	290 (65)	350 (78)	410 (92)	
			25	300 (67)	340 (76)	400 (90)	470 (105)	560 (126)	
			36	340 (76)	380 (85)	450 (101)	530 (119)	630 (141)	
			50	380 (85)	430 (96)	500 (112)	600 (135)	700 (157)	
	90 (3.54)	**ARL98**	5	380 (85)	420 (94)	470 (105)	540 (121)	630 (141)	600 (135)
			7.2	430 (96)	470 (105)	530 (119)	610 (137)	710 (159)	
			10	480 (108)	530 (119)	590 (132)	680 (153)	790 (177)	

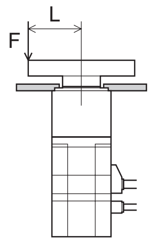
Type	Frame size [mm (in.)]	Unit model	Gear ratio	Permissible radial load [N (lb.)]					Permissible axial load [N (lb.)]
				Distance from the tip of motor output shaft					
				0 mm (0 in.)	5 mm (0.20 in.)	10 mm (0.39 in.)	15 mm (0.59 in.)	20 mm (0.79 in.)	
PS geared	90 (3.54)	**ARL98**	25	650 (146)	720 (162)	810 (182)	920 (200)	1070 (240)	600 (135)
			36	730 (164)	810 (182)	910 (200)	1040 (230)	1210 (270)	
			50	820 (184)	910 (200)	1020 (220)	1160 (260)	1350 (300)	
PN geared	42 (1.65)	**ARL46**	5	80 (18)	95 (21)	120 (27)	160 (36)	−	100 (22)
			7.2	90 (20)	110 (24)	130 (29)	180 (40)	−	
			10	100 (22)	120 (27)	150 (33)	200 (45)	−	
	60 (2.36)	**ARL66**	5	240 (54)	260 (58)	280 (63)	300 (67)	330 (74)	200 (45)
			7.2	270 (60)	290 (65)	310 (69)	340 (76)	370 (83)	
			10	300 (67)	320 (72)	350 (78)	380 (85)	410 (92)	
			25	410 (92)	440 (99)	470 (105)	520 (117)	560 (126)	
			36	360 (81)	410 (92)	480 (108)	570 (128)	640 (144)	
			50	360 (81)	410 (92)	480 (108)	570 (128)	700 (157)	
	90 (3.54)	**ARL98**	5	370 (83)	390 (87)	410 (92)	430 (96)	460 (103)	600 (135)
			7.2	410 (92)	440 (99)	460 (103)	490 (110)	520 (117)	
			10	460 (103)	490 (110)	520 (117)	550 (123)	580 (130)	
			25	630 (141)	660 (148)	700 (157)	740 (166)	790 (177)	
			36	710 (159)	750 (168)	790 (177)	840 (189)	900 (200)	
			50	790 (177)	840 (189)	890 (200)	940 (210)	1000 (220)	
Harmonic geared	42 (1.65)	**ARL46**	50, 100	180 (40)	220 (49)	270 (60)	360 (81)	510 (114)	220 (49)
	60 (2.36)	**ARL66**		320 (72)	370 (83)	440 (99)	550 (123)	720 (162)	450 (101)
	90 (3.54)	**ARL98**		1090 (240)	1150 (250)	1230 (270)	1310 (290)	1410 (310)	1300 (290)

Permissible moment load of the harmonic geared type

When installing an arm or table on the flange surface, calculate the moment load using the formula below if the flange surface receives any eccentric load. The moment load should not exceed the permissible value specified in the table below.

Moment load: M [N·m (oz-in)] = F × L

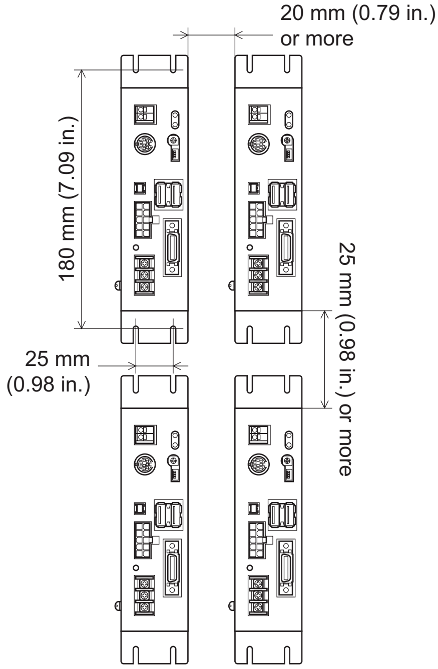
Unit mode	Permissible moment load [N·m (oz-in)]
ARL46	5.6 (790)
ARL66	11.6 (1640)



5.5 Installing the driver

When installing the driver in an enclosure, be sure to install vertically.

There must be a clearance of at least 25 mm (0.98 in.) in the horizontal and vertical directions, respectively, between the driver and enclosure or other equipment within the enclosure. When two or more drivers are to be installed side by side, provide 20 mm (0.79 in.) and 25 mm (0.98 in.) clearances in the horizontal and vertical directions, respectively.



Note

- Install the driver in an enclosure.
- Do not install any equipment that generates a large amount of heat or noise near the driver.
- Do not install the driver underneath the controller or other equipment vulnerable to heat.
- Check ventilation if the ambient temperature of the driver exceeds 40 °C (104 °F).
- Install the driver vertically.

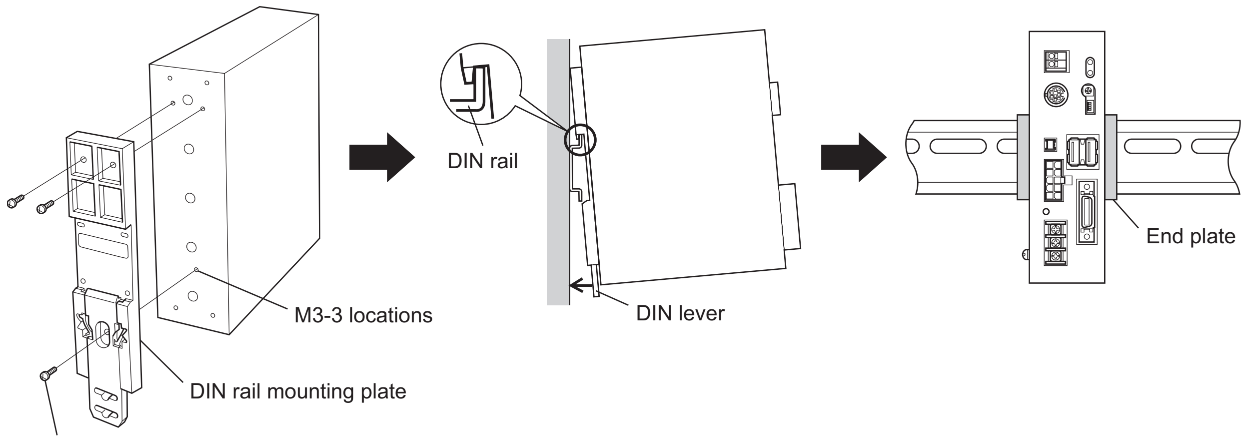
■ Installation method

To install a driver, attach it to a metal plate using driver mounting brackets or mount it to a DIN rail. In the presence of a great amount of vibration, do not use a DIN rail. Screw down the driver directly through the use of driver mounting brackets.

• Mounting to DIN rail

When mounting the driver to a DIN rail, use a optional DIN rail mounting plate (sold separately) and attach it to a 35 mm (1.38 in.) wide DIN rail.

1. Attach the DIN rail mounting plate to the rear panel of the driver with the screws.
   Screws are supplied with DIN rail mounting plate.
   Tightening torque: 0.3 to 0.4 N·m (42 to 56 oz-in)
2. Pulling the DIN lever downward, hook the tab of the DIN rail mounting plate on the DIN rail and push the driver until the DIN lever locks.
3. Secure the driver using end plates.

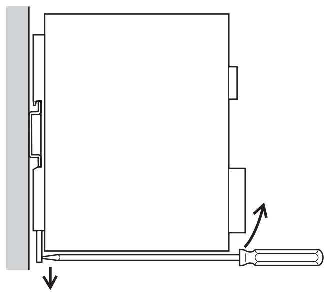


Note
- Do not use the mounting holes for the DIN rail mounting plate for any purpose other than securing the DIN rail mounting plate.
- Be sure to use the supplied screws when securing the DIN rail mounting plate. The use of screws that would penetrate 3 mm (0.118 in.) or more through the surface of the driver may cause damage to the driver.

• Removing from DIN rail

Pull the DIN lever down until it locks using a flat-head screwdriver, and lift the bottom of the driver to remove it from the rail.

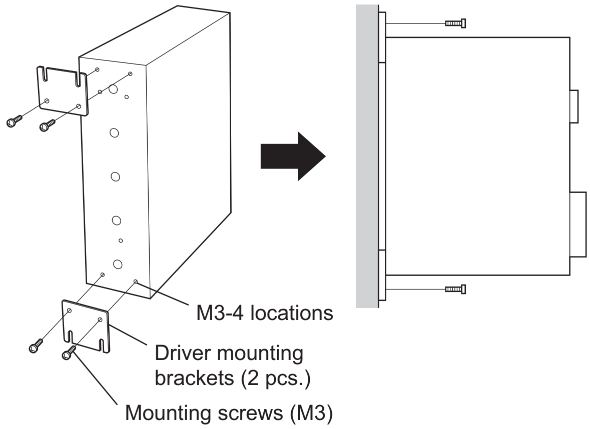
Note Use a force of about 10 to 20 N (2.2 to 4.5 lb.) to pull the DIN lever to lock it. Excessive force may damage the DIN lever.



- Using driver mounting brackets

Install the driver on a flat metal plate having excellent vibration resistance and heat conductivity.

1. Attach the driver mounting brackets to the back of the driver, using supplied screws for the brackets (M3, four pcs.)
   Tightening torque: 0.5 to 0.6 N·m (71 to 85 oz-in)
2. Install the driver by securing it with four bolts (not supplied) to the metal plate.
   Leave no gap between the driver and plate.



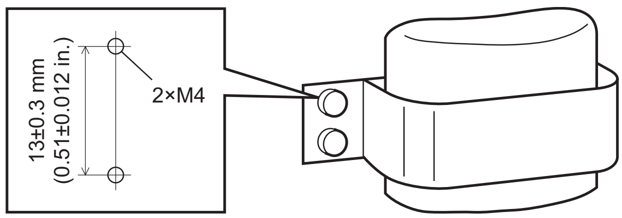
Note

- Do not use the mounting holes for the driver mounting brackets for any purpose other than securing the driver mounting brackets.
- Be sure to use the supplied screws when securing the driver mounting brackets. The use of screws that would penetrate 3 mm (0.118 in.) or more through the surface of the driver may cause damage to the driver.

5.6 Installing the battery (sold separately)

The battery has a built-in protective circuit. Use the optional battery holder to secure the battery.

Battery installation dimensions



5.7 Installing and wiring in compliance with EMC Directive

■ Introduction

EMC Directive

The **ARL** series has been designed and manufactured for incorporation in general industrial machinery. The EMC Directive requires that the equipment incorporating this product comply with these directives. The installation and wiring method for the motor and driver are the basic methods that would effectively allow the customer's equipment to be compliant with the EMC Directive.
This product has conducted EMC testing under the conditions specified in "Example of motor and driver installation and wiring" on p 21
The compliance with the EMC Directive of the entire equipment will vary depending on such factors as the configuration, wiring and installation condition of other control-system devices or electrical parts used together with this product. Therefore, be sure to confirm the compliance of the equipment in a finished condition that all parts including this product are installed in the equipment.

Applicable Standards

EMI	EN 55011 group 1 class A EN 61000-3-2 EN 61000-3-3
EMS	EN 61000-6-2 EN 61800-3

This type of PDS is not intended to be used on a low-voltage public network which supplies domestic premises; radio frequency interference is expected if used on such a network.

■ Installing and wiring in compliance with EMC Directive

Effective measures must be taken against the EMI that the motor and driver may give to adjacent control-system equipment, as well as the EMS of the motor and driver itself, in order to prevent a serious functional impediment in the machinery. The use of the following installation and wiring methods will enable the motor and driver to be compliant with the EMC directive. Refer to "CE Marking" on p.4 for the applicable standards.

• Connecting mains filter for power supply line

- Connect a mains filter in the AC input line to prevent the noise generated in the driver from propagating externally through the power supply line. Use a mains filter as FN2070-10-06 (Schaffner EMC) or HF2010A-UPF (SOSHIN ELECTRIC CO.,LTD), and install it as close to the driver as possible.
- Use a shielded cable of AWG18 (0.75 mm^2) or more for AC input cable and mains-filter output cable. Use cable clamps and other means to secure the cables firmly to the surface of the enclosure. Do not place the AC input cable parallel with the mains-filter output cable. Parallel placement will reduce mains-filter effectiveness if the enclosure's internal noise is directly coupled to the power supply cable by means of stray capacitance.
- Connect the ground terminal of the mains filter to the grounding point, using as thick and short a wire as possible.

• Connecting surge arrester

Use a surge arrester or equivalent as below.

Manufacturer	Single-phase 100-115 V	Single-phase 200-230 V
SOSHIN ELECTRIC CO.,LTD	LT-C12G801WS	
OKAYA ELECTRIC INDUSTRIES CO., LTD.	R·A·V-781BWZ-4	

Note When measuring dielectric strength of the equipment, be sure to remove the surge arrester, or the surge arrester may be damaged.

• Connecting reactor for AC power line

When inputting single-phase 230 V, insert a reactor (5 A, 2 mH) in the AC power line to ensure compliance with EN 61000-3-2.

• Power supply for electromagnetic brake (for electromagnetic brake motor only)

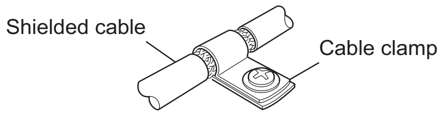
If an external DC power supply is required for the use of the electromagnetic brake, use a DC power supply that complies with the EMC Directive.
Use a shielded cable for wiring, and keep the wiring and grounding as short as possible.
See "Wiring the signal cable" for grounding the shielded cable.

• How to ground

The cable used to ground the driver, motor and mains filter must be as thick and short to the grounding point as possible so that no potential difference is generated. Choose a large, thick and uniformly conductive surface for the grounding point.
Use the Protective Earth Terminal for grounding the motor and driver. Refer to p.23 for the way to ground.

• Wiring the signal cable

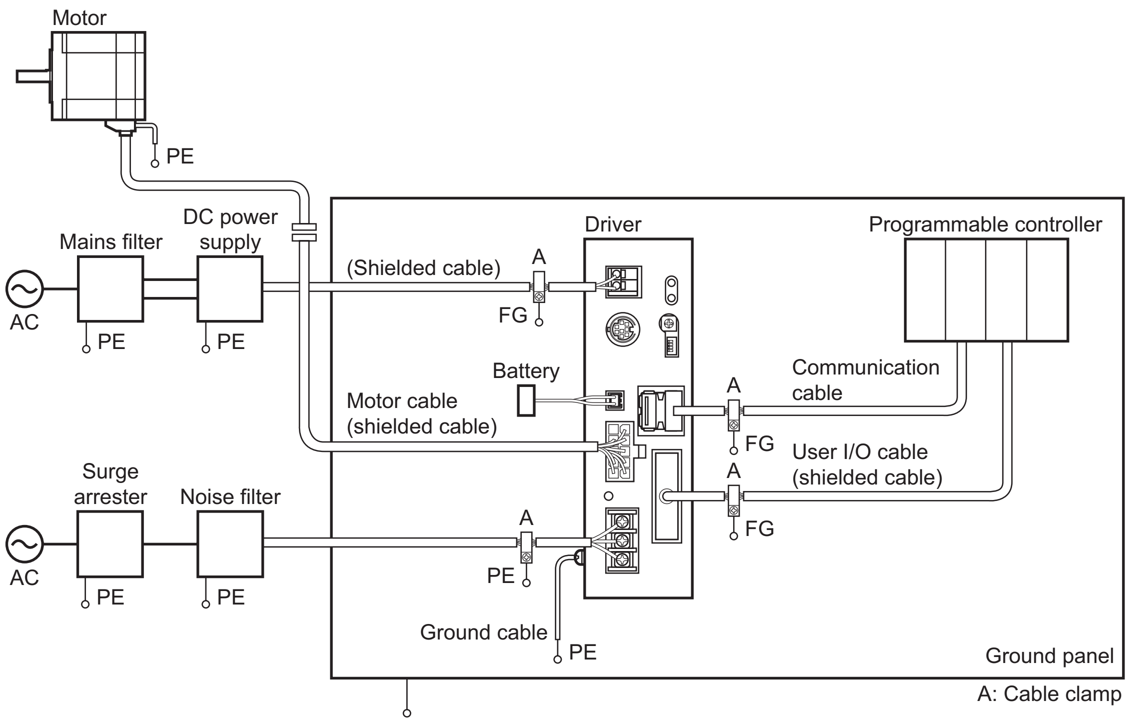
Use a shielded cable of AWG28 (0.08 mm^2) in diameter for the I/O cable, and keep it as short as possible.
To ground a shielded cable, use a metal clamp or similar device that will maintain contact with the entire circumference of the shielded cable. Attach a cable clamp as close to the end of the cable as possible, and connect it to an appropriate grounding point.



• Notes about installation and wiring

- Connect the motor, driver and other peripheral control equipment directly to the grounding point so as to prevent a potential difference from developing between grounds.
- When relays or electromagnetic switches are used together with the system, use mains filters and CR circuits to suppress surges generated by them.
- Keep cables as short as possible without coiling and bundling extra lengths.
- Place the power cables such as the motor and power supply cables as far apart [100 to 200 mm (3.94 to 7.87 in.)] as possible from the signal cables. If they have to cross, cross them at a right angle. Place the AC input cable and output cable of a noise filter separately from each other.
- When extending the distance between the motor and driver, it is recommended that an optional motor connection cable (sold separately) be used, since the EMC measures are conducted using the Oriental Motor motor connection cable.

- Example of motor and driver installation and wiring



A: Cable clamp

■ Precautions about static electricity

Static electricity may cause the driver to malfunction or suffer damage. While the driver is receiving power, handle the driver with care and do not come near or touch the driver.
Always use an insulated screwdriver to adjust the driver's switches.

Note The driver uses parts that are sensitive to electrostatic charge. Before touching the driver, turn off the power to prevent electrostatic charge from generating. If an electrostatic charge is impressed on the driver, the driver may be damaged.

6 Connection

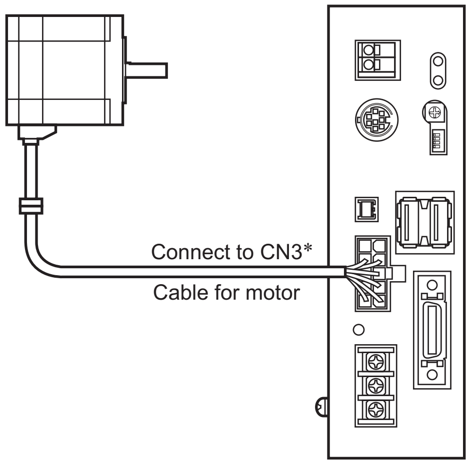
This chapter explains how to connect/ground the motor, driver, power supply, control I/O, or communication cable.

6.1 Connecting the motor

Connect the motor cable and supplied cable for motor, and connect to the motor connector (CN3).

Note

- Have the connector plugged in securely. Insecure connector connection may cause malfunction or damage to the motor or driver.
- To disconnect the plug, pull the plug while using the fingers to press the latches on the plug.
- When plugging/unplugging the connector, turn off the power and wait for 10 seconds or more. The residual voltage may cause electric shock.
- When installing the motor on a moving part, use an optional flexible cable offering excellent flexibility. Refer to p.73 for optional flexible cable.

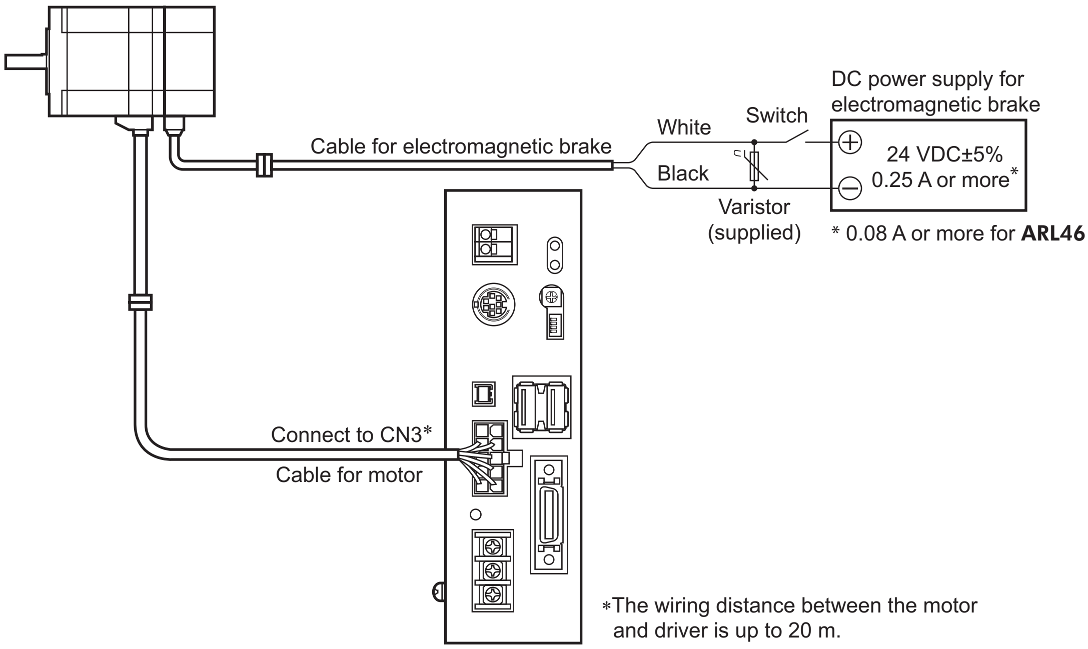


*The wiring distance between the motor and driver is up to 20 m.

6.2 Connecting an electromagnetic brake motor

For the electromagnetic brake, provide a DC power supply of 24 VDC ±5%, 0.25 A (**ARL46**: 0.08 A) or more. Use a shielded cable of AWG24 (0.2 mm^2) or more to extend the electromagnetic brake to the DC power supply, keeping the length as short as possible.

1. Connect the motor cable and supplied cable for motor, and connect to the motor connector (CN3).
2. Connect the electromagnetic brake cable and supplied cable for electromagnetic brake, and connect the lead wires to the DC power supply.
   Connect white lead wire to +24 VDC terminal of the DC power supply.
   Connect black lead wire to GND terminal of the DC power supply.
3. Connect the varistor (supplied) in parallel between the +24VDC terminal and the GND terminal of the DC power supply.



*The wiring distance between the motor and driver is up to 20 m.

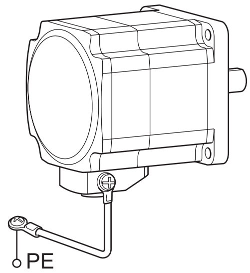
Note
- Do not use the electromagnetic brake beyond their specified values. Applying a voltage over the specification will increase the temperature rise in the electromagnetic brake and may damage the motor. Conversely, insufficient voltage may prevent the brake from releasing.
- Be sure to connect the varistor (supplied) to protect the switch contacts and prevent noise.
- The lead wires for the electromagnetic brake are polarized. Connecting the lead wires in reversed polarity will not properly operate the electromagnetic brake.
- Provide separate power supplies for the I/O signals and the electromagnetic brake.

6.3 Grounding the motor and driver

■ Grounding the motor

Be sure to ground the Protective Earth Terminal.
Tightening torque: 1.2 N·m (170 oz-in)

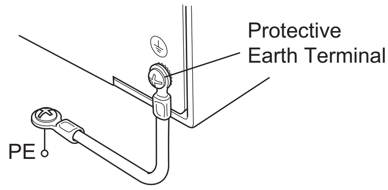
Use a grounding wire of AWG18 (0.75 mm^2) or more.
Use a round crimp terminal in combination with a crow washer and bolt it in place to secure the grounding connection. The motor does not come with a protective earth cable or crimp terminal.



■ Grounding the driver

Be sure to ground the Protective Earth Terminal (screw size: M4) located on the driver side.
Tightening torque: 0.5 to 0.6 N·m (71 to 85 oz-in)

Use a grounding wire larger than AWG18 (0.75 mm^2), and do not share the protective earth terminal with a welder or any other power equipment. When grounding the Protective Earth Terminal, use a round terminal and affix the grounding point near the driver.



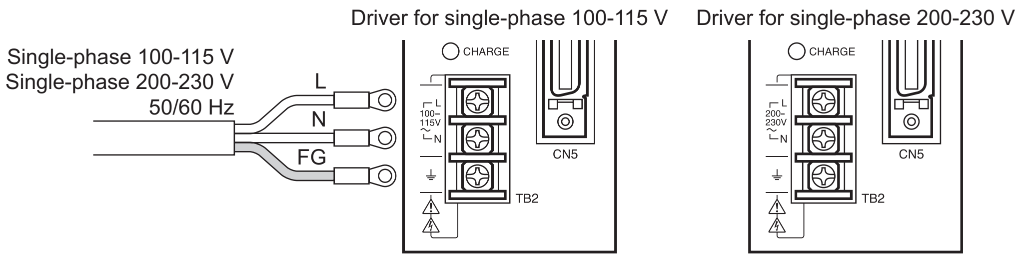
6.4 Connecting the main power supply

Connect the power supply cable to the power supply terminal (TB2).

Note
- Furnish a power supply capable of supplying adequate driver input current. If the current capacity is insufficient, the transformer may be damaged or the torque may drop causing the motor to malfunction.
- Do not wire the power supply cable of the driver in the same cable duct with other power line or motor cable. Doing so may cause malfunction due to noise.
- When plugging/unplugging the connector, turn off the power and wait for 10 seconds or more. The residual voltage may cause electric shock.
- When connecting the main power supply, turn off the power and wait for the CHARGE LED to turn off (approximately 6 minutes). The residual voltage may cause electric shock.

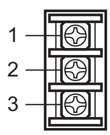
■ Connection method

Connect the live side of the power supply to the L terminal of main power supply input terminals and the neutral side to the N terminal. Connect the FG terminal to the grounding point of the power supply.



Pin assignment of main power supply input terminals (TB2)

Pin No.	Signal name	Description
1	L	Main power supply input
2	N	
3	⏚	Frame Ground



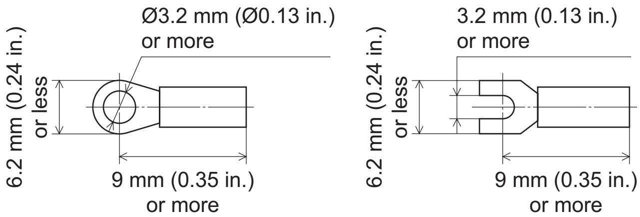
Use a power supply capable of supplying the current capacity as shown below.

Motor model	Single-phase 100-115 V	Single-phase 200-230 V
ARLM46	2.7 A or more	1.7 A or more
ARLM66	4.1 A or more	2.4 A or more
ARLM69	6.2 A or more	3.5 A or more
ARLM98	5.6 A or more	3.4 A or more
ARLM911	6.1 A or more	4.1 A or more

■ Screw size of main power supply input terminal and applicable lead wire

- Terminal screw size: M3
- Tightening torque: 0.8 N·m (113 oz-in)
- Minimum applicable lead wire: AWG16 to 18 (0.75 to 1.25 mm^2)

Applicable crimp terminal

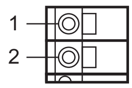


6.5 Connecting the control power supply

Connect the control power supply to the control power supply input terminal (TB1).

- Pin assignment of control power supply input terminals (TB1)

Pin No.	Signal name	Description
1	+	Control power supply input 24 VDC±10% 1.0 A
2	−	Control power supply GND



- Applicable lead wire
  - Applicable lead wire : AWG22 to 16 (0.3 to 1.25 mm^2)
  - Bare wire length: 11 mm (0.43 in.)

6.6 Connecting/charging/disconnecting the battery

If the driver is of the absolute type, connect the optional battery (sold separately) to the driver.
The product is shipped with an uncharged battery. Follow the procedure below when charging the battery for the first time:

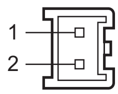
■ Connecting/charging method

1. Connect the battery to the driver's battery connector (CN2).
2. Supply the control power to the driver.
   The battery starts charging. It takes approx. 48 hours to fully charge the battery [at an ambient temperature of 20 °C (68 °F)].

Note
- The product is shipped with an uncharged battery. Charge the battery for the first time.
- Data can be backed up with a charged battery for a period of up to 15 days. Disconnect the battery if the driver is not to be powered up for at least 15 days. Failure to do so may cause battery fluid leakage and/or decreased performance.

Pin assignment of battery connector (CN2)

Pin No.	Signal name	Description
1	+	Battery power supply input
2	GND	Battery power supply GND



■ Disconnecting method

Cut off the main power supply, and disconnect the battery from the driver.

Note
- Be sure to cut off the main power supply before disconnect the battery.
- Once the battery is disconnected, the current position will be reset. Be sure to perform return-to-home operation after replacing the batteries.

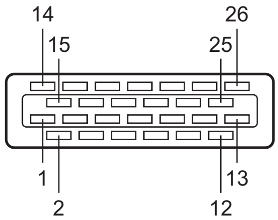
6.7 Connecting the user I/O

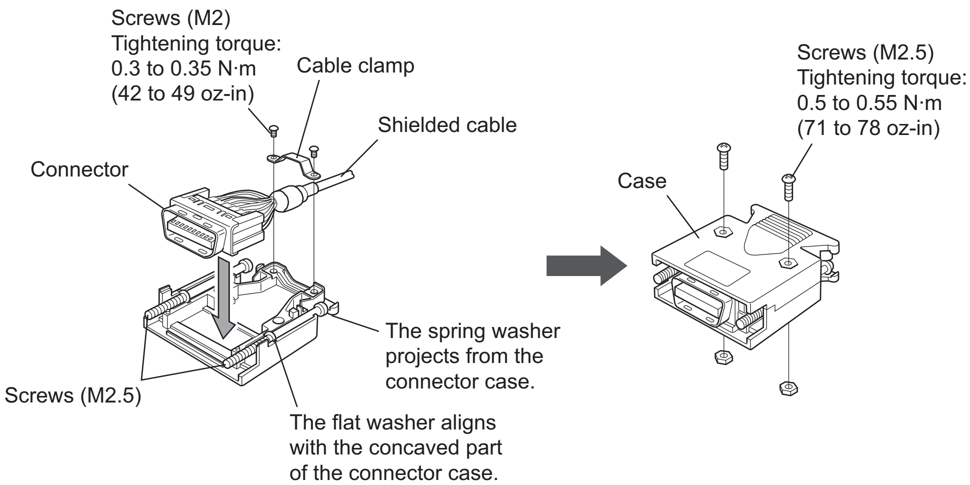
Use the supplied CN5 connector (26 pins) to connect the user I/O.

■ Assembling the user I/O connector

1. Solder the shielded cable to the connector pins and assembling the connector.
   Shielded cable is not supplied. Use a shielded cable with a size of AWG28 to 26 (0.08 to 0.14 mm^2) or more.

Connector pin assignments
(Viewed from the soldering side)





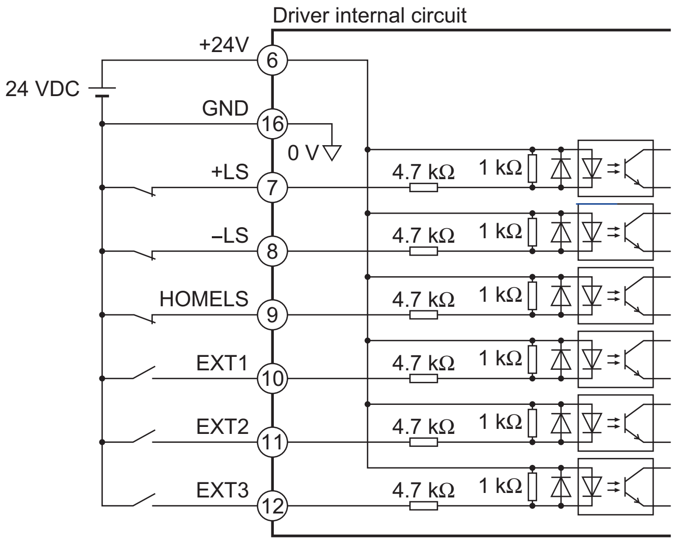
2. Connect the installed connector to the driver's user I/O connector (CN5) and tighten the screws.
   Tightening torque: 0.3 to 0.35 N·m (42 to 49 oz-in)

■ Pin assignment of user I/O connector (CN5)

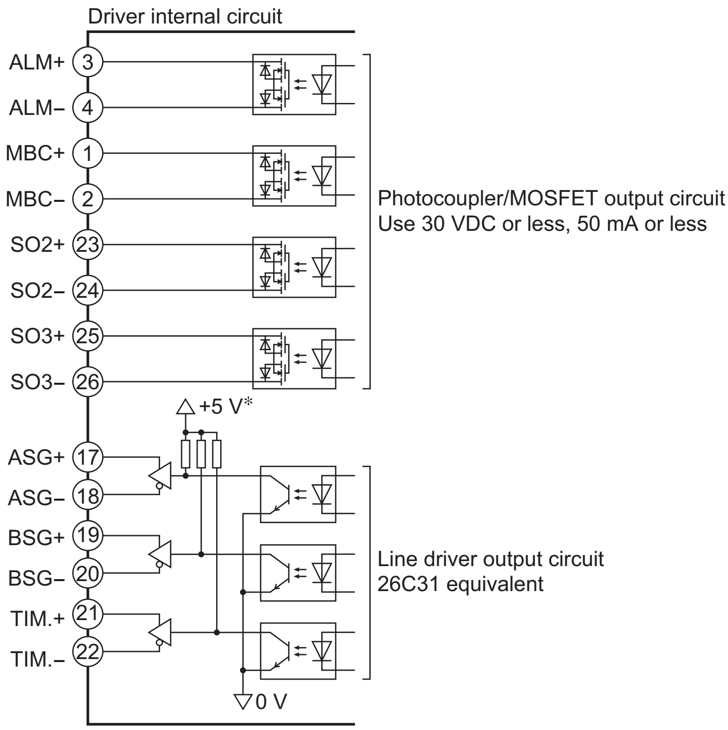
Pin No.	Signal name	Description	Direction
1	MBC+	Electromagnetic brake control	Output
2	MBC−		
3	ALM+	Alarm output	
4	ALM−		
5	NC	Not used	−
6	+24 V	Control power supply input for I/O	Input
7	+LS	+limit sensor	
8	−LS	−limit sensor	
9	HOMELS	HOME sensor	
10	EXT1	External latch signal 1	
11	EXT2	External latch signal 2	
12	EXT3	External latch signal 3	
13	NC	Not used	−

Pin No.	Signal name	Description	Direction
14	NC	Not used	−
15	NC		
16	GND	Power supply GND for I/O	Input
17	ASG+	A-phase output (Line-driver)	Output
18	ASG−		
19	BSG+	B-phase output (Line-driver)	
20	BSG−		
21	TIM.+	Timing signal output (Line-driver)	
22	TIM.−		
23	SO2+	General output 2	
24	SO2−		
25	SO3+	General output 3	
26	SO3−		

- Internal input circuit

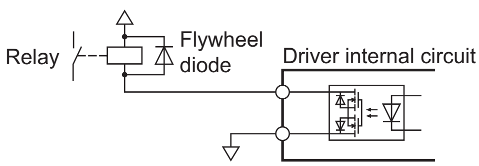


- Internal output circuit



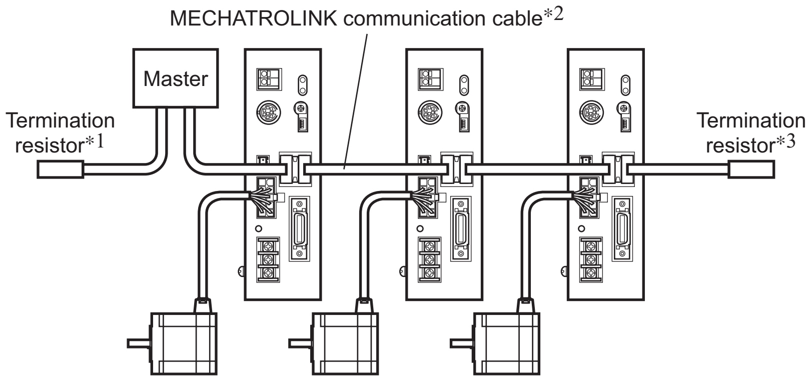
* The +5 V power is generated from the control power supply (+24 V).

Note If a relay or other inductive load is connected to the transistor/open-collector output circuit, connect a flywheel diode as a countermeasure for counter voltage that may generate due to the inductive load. When selecting a diode, check the recommended values for the applicable relay.



6.8 Connecting the MECHATROLINK communication

Connect the MECHATROLINK communication cable to the MECHATROLINK communication connector (CN4). You can use the vacant connectors to connect a different driver.

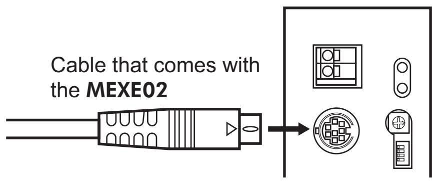


*1 The master may come with a built-in termination resistor that functions as the termination resistor on master side. For details, refer to the operating manual for the master.
*2 For the MECHATROLINK communication cable, use a dedicated MECHATROLINK communication cable manufactured by YASKAWA Electric Corporation.
*3 For the termination resistor on slave side, use a dedicated termination resistor for MECHATROLINK communication, manufactured by YASKAWA Electric Corporation.

6.9 Connecting the communication cable

Connect the cable that comes with the **MEXE02** to the communication connector (CN1) on the driver.

Note When grounding the positive terminal of the power supply, do not connect any equipment (PC, etc.) whose negative terminal is grounded. Doing so may cause the driver and these equipment to short, damaging both.

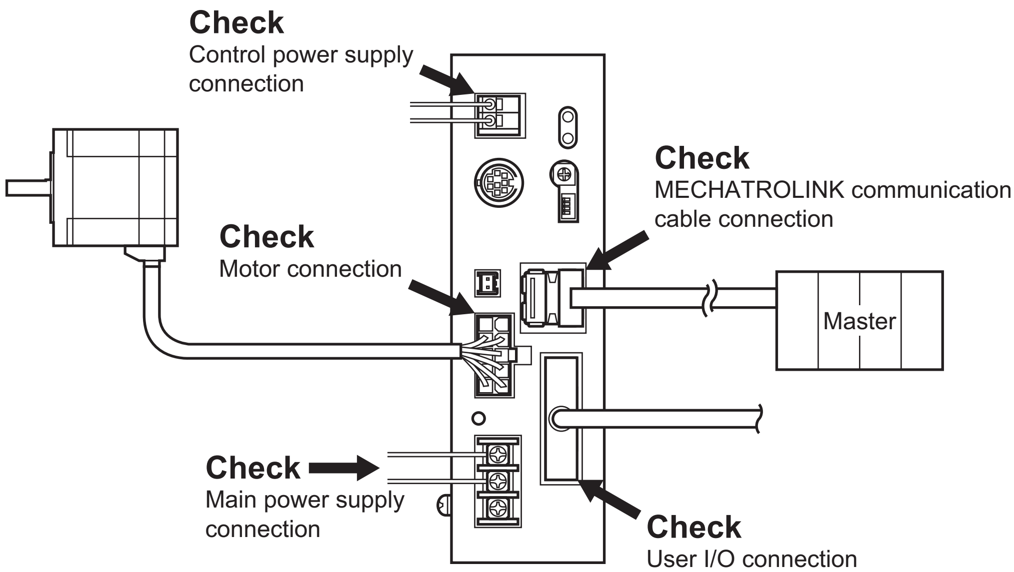


7 Guidance

If you are new to the **ARL** series MECHATROLINK, read this section to understand the operating methods along with the operation flow.

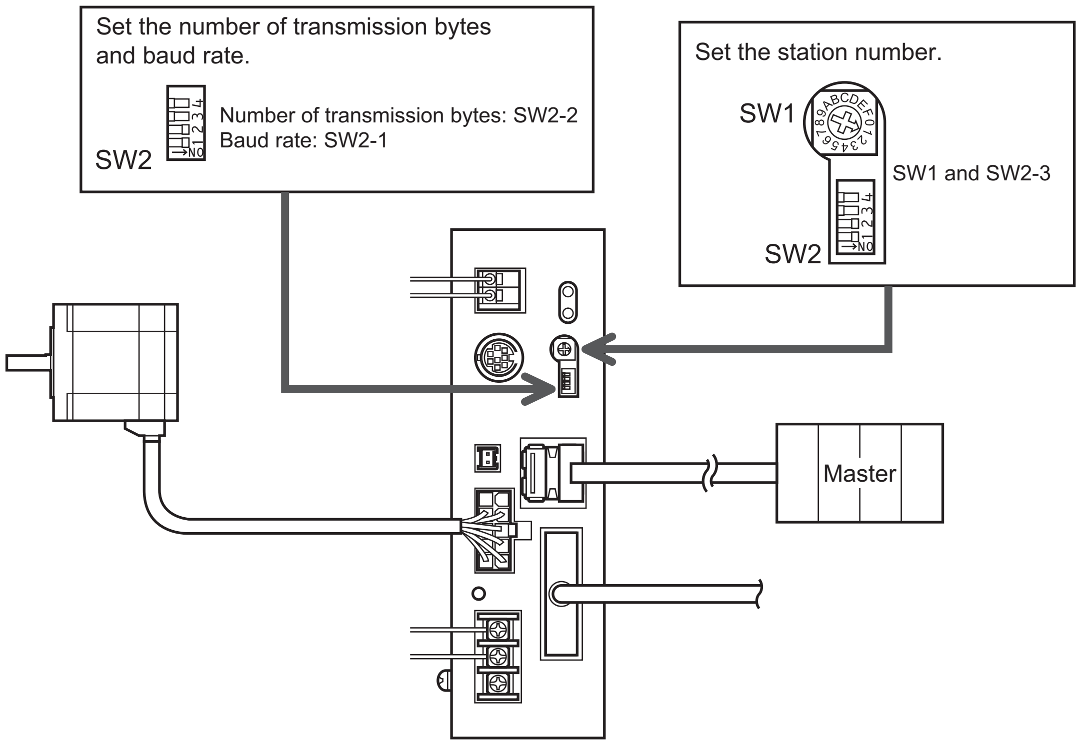
Note Before operating the motor, check the condition of the surrounding area to ensure safety.

STEP 1 Check the connection

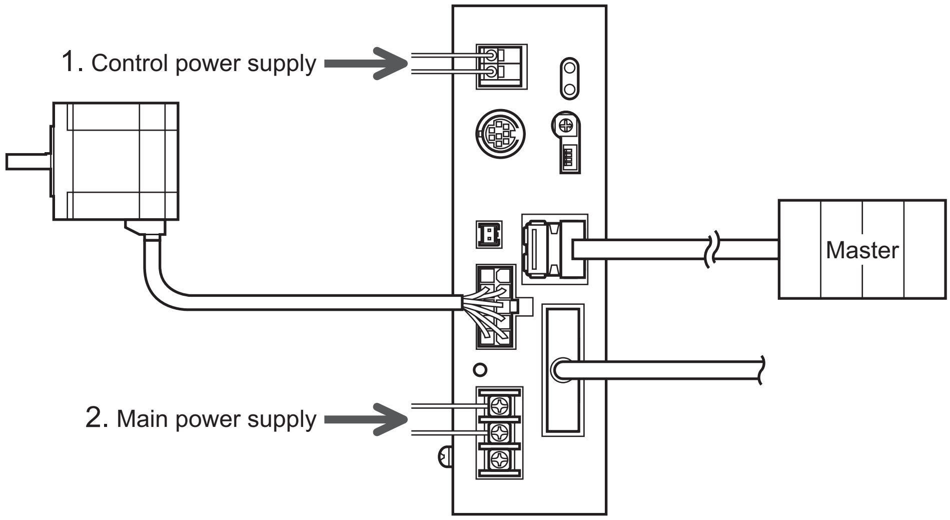


Note The +LS input, −LS input and HOMELS input are normally closed (contact B).

STEP 2 Set the switches



STEP 3 Turn on the power



STEP 4 Set the master device

- If a PLC master manufactured by YASKAWA Electric Corporation is used, set the model type to "Stepping Motor DRV."
- If a master manufactured by KEYENCE CORPORATION is used, select "Stepping" from the list of MECHATROLINK-II slaves and set it as the model type.
- Set the MECHATROLINK- II communication period.
  Set the communication period for 0.5. to 3.0 ms (0.5 ms unit).

STEP 5 Operate the motor

1. Issue a SV_ON command to excite the motor.
2. Set the target speed and issue a FEED command.
   The motor runs continuously in the positive direction.

STEP 6 Were you able to operate the motor properly?

How did it go? Were you able to operate the motor properly?
If the motor does not function, check the following points:

- Is any alarm present?
  Is the OP LED lit red?
- Are the main power supply, control power supply, motor and MECHATROLINK communication cable connected securely?
- Is the main power supply on?
- Have you issued a SV_ON command?
- Are the Station-number setting switch, MECHATROLINK communication setting switch and communication period set correctly?
- Is a termination resistor connected to both sides of the MECHATROLINK communication line?
- Are the +LS input, −LS input and HOMELS input connected securely? Is the logic of these inputs set correctly?
- Have you turned on the user I/O power?

8 Setting the MECHATROLINK-Ⅱ communication

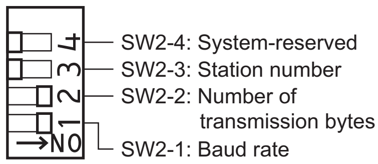
Note

- Be sure to turn off the driver power before setting these switches. If the switches are set while the driver power is on, the setting will not become effective.
- Do not turn off the driver power while data is being written to the NV memory. Doing so may damage the data.

8.1 Setting the communication specifications

Set the MECHATROLINK-Ⅱ communication specifications using the MECHATROLINK communication setting switch (SW2).
When changing the switch, cycle the control power supply.

SW2-4	System-reserved	Keep this switch in the OFF position.
SW2-3	Station number	OFF: Station address=40h+SW1 (factory setting) ON: Station address=50h+SW1
SW2-2	Number of transmission bytes	OFF: 17 bytes ON: 32 bytes (factory setting)
SW2-1	Baud rate	OFF: 4 Mbps ON: 10 Mbps (factory setting)



Note

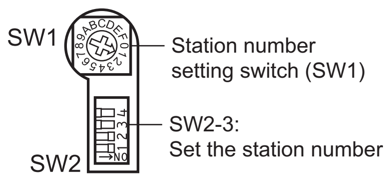
- Turn OFF all bits when connecting to the MECHATROLINK network.
- The baud rate setting of 4 Mbps cannot be combined with the number of transmission bytes setting of 32 bytes (SW2-1 is OFF and SW2-2 is ON).
- If the baud rate is set to 4 Mbps and number of transmission bytes to 17 bytes (SW2-1 is OFF and SW2-2 is OFF), MECHATROLINK-II communication cannot be established.
- MECHATROLINK-Ⅱ communication period is 0.5 to 3 ms (0.5 ms unit).
  MECHATROLINK communication period is 2 ms (fixed).

8.2 Setting the station number

Set the driver's station number using the station number setting switch (SW1) and MECHATROLINK communication setting switch (SW2-3).
When changing the switch, cycle the control power supply.
Factory setting: 41h (SW2-3: OFF, SW1: 1)

SW2-3	SW1	Station number
OFF	0	Disable
	1	41h
	2	42h
	3	43h
	4	44h
	5	45h
	6	46h
	7	47h
	8	48h
	9	49h
	A	4Ah
	B	4Bh
	C	4Ch
	D	4Dh
	E	4Eh
	F	4Fh

SW2-3	SW1	Station number
ON	0	50h
	1	51h
	2	52h
	3	53h
	4	54h
	5	55h
	6	56h
	7	57h
	8	58h
	9	59h
	A	5Ah
	B	5Bh
	C	5Ch
	D	5Dh
	E	5Eh
	F	5Fh



9 Operation via MECHATROLINK-Ⅱ communication

9.1 Commands for MECHATROLINK-Ⅱ

This section describes the commands of **ARL** series MECHATROLINK-Ⅱ driver.

How to read the table	Processing category	N: Network command D: Data communication command C: Control command M: Motion command
	Synchronization category	A: Asynchronous command S: Synchronous command

■ List of main commands

Main commands support the MECHATROLINK communication and the MECHATROLINK-Ⅱ communication.

Command code	Name	Command name	Processing category	Synchronization category	Ref.
00h	NOP	Invalid command	N	A	p.32
01h	PRM_RD	Parameter read	D	A	p.32
02h	PRM_WR	Parameter write	D	A	p.33
03h	ID_RD	ID read	D	A	p.33
04h	CONFIG	Configuration setup request	C	A	p.34
05h	ALM_RD	Alarm/warning read	D	A	p.34
06h	ALM_CLR	Alarm/warning clear	C	A	p.34
0Dh	SYNC_SET	Synchronization request	N	A	p.35
0Eh	CONNECT	MECHATROLINK-II connection request	N	A	p.35
0Fh	DISCONNECT	Disconnection request	N	A	p.36
1Bh	PPRM_RD	Nonvolatile parameter read	Not implemented	Not implemented	−
1Ch	PPRM_WR	Nonvolatile parameter write	D	A	p.37
20h	POS_SET	Coordinate system setting	D	A	p.37
21h	BRK_ON	Brake actuation request	C	A	p.38
22h	BRK_OFF	Brake release request	C	A	p.38
23h	SENS_ON	Sensor ON	C	A	p.38
24h	SENS_OFF	Sensor OFF	C	A	p.39
25h	HOLD	Motion stop request	M	A	p.39
30h	SMON	Servo status monitor	D	A	p.40
31h	SV_ON	Servo ON	C	A	p.40
32h	SV_OFF	Servo OFF	C	A	p.41
34h	INTERPOLATE	Interpolation feed	M	S	p.41
35h	POSING	Positioning	M	A	p.42
36h	FEED	Constant-speed feed	M	A	p.42
38h	LATCH	Interpolation feed with position latch function	M	S	p.43
39h	EX_POSING	External input positioning	M	A	p.44
3Ah	ZRET	Return-to-home	M	A	p.45
0Eh	CONNECT	MECHATROLINK-I connection request	N	A	p.47

■ List of sub commands

Sub commands support the MECHATROLINK-Ⅱ communication.

Command code	Name	Command name	Processing category	Synchronization category	Ref.
00	NOP	Invalid command	N	A	p.48
01	PRM_RD	Parameter read	D	A	p.48
02	PRM_WR	Parameter write	D	A	p.48
05	ALM_RD	Alarm/warning read	D	A	p.48
1C	PPRM_WR	Nonvolatile parameter write	D	A	p.49
30	SMON	Servo status monitor	D	A	p.49

9.2 Main command specifications

MECHATROLINK-Ⅱ main commands have upward compatibility with MECHATROLINK commands.
These commands consist of bytes 1 to 31 of command/response data.

■ Invalid command (NOP: 00h)

Byte	Command	Response
1	NOP (00h)	NOP (00h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 15	0	0
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Network command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	Invalid command or no command. The current status (ALARM, STATUS) is returned from the slave as a response.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (NOP), and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Parameter read command (PRM_RD: 01h)

Byte	Command	Response
1	PRM_RD (01h)	PRM_RD (01h)
2	0	ALARM
3	0	STATUS
4	0	
5	No.	No.
6		
7	SIZE	SIZE
8 to 15	0	PARAMETER
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Specify the parameter number and data size and read the parameter. No.: parameter number SIZE: parameter size PARAMETER: parameter setting value (little endian) For details on parameter, refer on p.55. • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the specified number is outside the setting range (03h) · When the value of SIZE does not match (03h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (PRM_RD), bit 2 (CMDRDY=1) of the status field (p.51), No. and SIZE, in the response.

■ Parameter write command (PRM_WR: 02h)

Byte	Command	Response
1	PRM_WR (02h)	PRM_WR (02h)
2	0	ALARM
3	0	STATUS
4	0	
5	No.	No.
6		
7	SIZE	SIZE
8 to 15	PARAMETER	PARAMETER
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	150 ms or less
Subcommand expansion	Refer to p.50.
Description	• Specify the parameter number and data size and write the parameter to the RAM. No.: parameter number SIZE: parameter size PARAMETER: parameter setting value (little endian) For details on parameter, refer on p.55. • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the specified number is outside the setting range (03h) · When the value of SIZE does not match (03h) · When the PARAMETER is outside the specified range (03h). · When a dedicated READ parameter is specified (03h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (PRM_WR), bit 2 (CMDRDY=1) of the status field (p.51), ALARM, No. , SIZE and PARAMETER, in the response.

■ ID read command (ID_RD: 03h)

Byte	Command	Response
1	ID_RD (03h)	ID_RD (03h)
2	0	ALARM
3	0	STATUS
4	0	
5	DEVICE_CODE	DEVICE_CODE
6	OFFSET	OFFSET
7	SIZE	SIZE
8 to 15	0	ID
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	Read the product information as ID data. DEVICE_CODE: see table below. OFFSET, SIZE: Specify the data obtained by DEVICE_CODE. 0≦OFFSET≦255、0≦SIZE≦8 ID: ID data is set in sequence, starting from byte 8, for the number of bytes specified by SIZE.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (ID_RD), bit 2 (CMDRDY=1) of the status field (p.51), DEVICE_CODE, OFFSET and SIZE, in the response.

• DEVICE_CODE list

Values in blank fields are indeterminable.

Type/Name	DEVICE_CODE	OFFSET	Content of ID															
			00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Product type	00h	00h	S	T	P	2	,	A	R	L	D	00h						
Software version	02h	00h	□	□	00h													
Vender code	0Fh	00h	05h	00h														
		10h	O	R	I	E	N	T	A	L	M	O	T	O	R	00h		

Product type : The model number of the driver is returned as an ASCII code.
Software version : The software version of the driver is returned as a binary code. (little endian).
Example) If the version is 1.02, “0201” is returned.

• Example of read

If DEVICE_CODE=00h, OFFSET=00h and SIZE=08h, “STP2, ARLD” will be acquired into the ID field.

■ Configuration setup request command (CONFIG: 04h)

Byte	Command	Response
1	CONFIG (04h)	CONFIG (04h)
2	0	ALARM
3	0	STATUS
4	0	
5	CFG_MODE	CFG_MODE
6 to 15	0	0
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	More than communication period
Subcommand expansion	Refer to p.50.
Description	If this command is executed, the motor will stop immediately and the servo will turn off. The electromagnetic brake will remain effective. CFG_MODE: fixed to 0
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (CONFIG) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Alarm/warning read command (ALM_RD: 05h)

Byte	Command	Response
1	ALM_RD (05h)	ALM_RD (05h)
2	0	ALARM
3	0	STATUS
4	0	
5	ALM_RD_MODE	ALM_RD_MODE
6 to 15	0	ALM_DATA (new) ALM_DATA (old)
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	See table below.
Subcommand expansion	Refer to p.50.
Description	• Read each present alarm/warning using an alarm code. ALM_RD_MODE: See table below. ALM_DATA: New alarms are set in sequence, starting from byte 6. • If the following conditions applies, a warning will generate and the command will be ignored: · When the ALM_RD_MODE is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (ALM_RD) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

• ALM_RD_MODE list

ALM_RD_MODE	Description	Processing time
0	Read the alarm and warning currently present.	Within the communication period
1	Read alarm records (up to 10).	150 ms or less

■ Alarm/warning clear command (ALM_CLR: 06h)

Byte	Command	Response
1	ALM_CLR (06h)	ALM_CLR (06h)
2	0	ALARM
3	0	STATUS
4	0	
5	ALM_CLR_MOD	ALM_CLR_MOD
6 to 15	0	0
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	See table below.
Subcommand expansion	Refer to p.50.
Description	• Resets the alarms and warnings that are present. Also clear the alarm records. ALM_CLR_MOD: Refer to next table. • If the following conditions applies, a warning will generate and the command will be ignored: · When the ALM_CLR_MOD is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (ALM_CLR) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

- ALM_CLR_MOD list

ALM_CLR_MOD	Description	Processing time
0	Resets the alarms and warnings that are present.	150 ms or less
1	Clear the alarm records.	250 ms or less

■ Synchronization request command (SYNC_SET: 0Dh)

Byte	Command	Response
1	SYNC_SET (0Dh)	SYNC_SET (0Dh)
2	0	ALARM
3	0	STATUS
4	0	
5 to 15	0	0
16	WDT	RWDT

Processing category	Network command
Synchronization category	Asynchronous command
Processing time	More than communication period
Subcommand expansion	Refer to p.50.
Description	Start synchronous communication. Specifically, the system switches from phase 2 (asynchronous communication) to phase 3 (synchronous communication). Synchronization is to be established after confirmation of a change in WDT (+1). If this command is received in phase 2 while the servo is on, the motor will stop immediately and the servo will turn off. The electromagnetic brake will remain effective. This command is ignored in phase 3 (no warning will generate).
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SYNC_SET) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

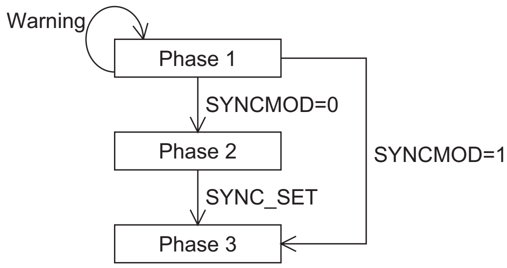
■ MECHATROLINK-Ⅱ connection request command (CONNECT: 0Eh)

Byte	Command	Response
1	CONNECT (0Eh)	CONNECT (0Eh)
2	0	ALARM
3	0	STATUS
4	0	
5	VER	VER
6	COM_MODE	COM_MODE
7	COM_TIME	COM_TIME
8 to 15	0	0
16	WDT	RWDT

Processing category	Network command
Synchronization category	Asynchronous command
Processing time	More than communication period
Subcommand expansion	Refer to p.50.
Description	• Request MECHATROLINK connection to be established. Upon completion of this command, the system will switch to phase 2 or 3 and communication with the master station will be enabled. Set the communication mode and communication period. VER: Set the version VER=21H (Ver.2.1) COM_MODE: Communication mode (see table below.) COM_TIME: Communication period For the communication period, set an integer multiple of the transmission period. To be specific, COM_TIME must be a multiple of 0.5 ms where the relationship “0.5 ms≦Transmission period × COM_TIME≦3 ms” is satisfied. • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the COM_MODE is outside the specified range (03h). · When the COM_TIME is outside the specified range (03h). · When the number of transmission bytes is 17 and SUBCMD is 1 (03h) · When the MECHATROLINK communication mode is active and VER is not 21H (03h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (CONNECT), bit 2 (CMDRDY=1) of the status field (p.51), and set data (VER, COM_MODE, COM_TIME), in the response.

- COM_MODE list

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
SUBCMD	0	0	0	DTMOD		SYNCMOD	0

Name	Description	Setting
SYNCMOD	Synchronization setting	0: Perform asynchronous communication (switch to phase 2). Watchdog check is disabled. Synchronous communication commands cannot be used. 1: Perform synchronous communication (switch to phase 3). Watchdog check is enabled. Synchronous communication commands can be used. 
DTMOD	Communication Method	00: Single transmission communication
SUBCMD	Subcommand setting	0: Subcommand disabled 1: Subcommand enabled

■ Disconnection request command (DISCONNECT: 0Fh)

Byte	Command	Response
1	DISCONNECT (0Fh)	DISCONNECT (0Fh)
2	0	ALARM
3	0	STATUS
4	0	
5 to 15	0	0
16	WDT	RWDT

Processing category	Network command
Synchronization category	Asynchronous command
Processing time	More than the communication period
Subcommand expansion	Refer to p.50.
Description	Request MECHATROLINK connection to be released. Upon completion of command the system will switch to phase 1. When this command is received, the following operations will be performed: · The motor will stop immediately and the servo will turn off. The electromagnetic brake will remain effective. · Reference point setting will be disabled (incremental specification only). · Position information will be initialized (incremental specification only). · The alarm is reset.
Confirmation of completion (Process performed by the master)	The slave returns a response. If completion is not confirmed, this command will be sent for at least two communication periods. Completion is confirmed based on byte 1 (DISCONNECT) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Nonvolatile parameter write command (PPRM_WR: 1Ch)

Byte	Command	Response
1	PPRM_WR (1Ch)	PPRM_WR (1Ch)
2	0	ALARM
3	0	STATUS
4	0	
5	No.	No.
6		
7	SIZE	SIZE
8 to 15	PARAMETER	PARAMETER
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	150 ms or less
Subcommand expansion	Refer to p.50.
Description	• Specify the parameter number, data size and parameter data and write the data to the NV memory and RAM. No.: parameter number SIZE: parameter size PARAMETER：parameter setting value (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the specified number is outside the setting range (03h) · When the value of SIZE does not match (03h) · When the PARAMETER is outside the specified range (03h). ·When a dedicated READ parameter is specified (03h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (PPRM_WR), bit 2 (CMDRDY=1) of the status field (p.51), No., SIZE and PARAMETER in the response.

Note The NV memory can be rewritten approx. 100,000 times.

■ Coordinate system setting command (POS_SET: 20h)

Byte	Command	Response
1	POS_SET (20h)	POS_SET (20h)
2	0	ALARM
3	0	STATUS
4	0	
5	PS_SUBCMD	PS_SUBCMD
6 to 9	POS_DATA	POS_DATA
10 to 15	0	0
16	WDT	RWDT

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	150 ms or less
Subcommand expansion	Refer to p.50.
Description	• Set the coordinate system. Once the reference point is set, the home position (ZPOINT) and soft limits will become effective. PS_SUBCMD: See table below. POS_DATA: Set value of reference point (little endian) −2,147,483,648≦POS_DATA≦+2,147,483,647 (Absolute position: −8,388,608≦POS_DATA≦ +8,388,607) • If the following condition applies, a warning will generate and the command will be ignored: · When the number set in PS_SUBCMD is outside the setting range (03h). • For the absolute specification to be used, the home position is written to the NV memory.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (POS_SET), bit 2 (CMDRDY=1) of the status field (p.51), PS_SUBCMD and POS_DATA, in the response.

• PS_SUBCMD list

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
REFE	0	0	0	POS_SEL			

Name	Description	Setting
POS_SEL	Coordinate system selection	3 (POS): Set the coordinate system in POS_DATA.
REFE	Reference point setting	1: Confirm the coordinate system. The home position (ZPOINT) and soft limits become effective.

■ Brake actuation request command (BRK_ON: 21h)

Byte	Command	Response
1	BRK_ON (21h)	BRK_ON (21h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	1 ms or less
Subcommand expansion	Refer to p.50.
Description	• Turn OFF the MBC output. Note that this command is valid when memory switch 4 (brake operation setting) is set to 0. MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When bit 9 of memory switch 4 is set to 1 (02h) · When the motor is currently excited (02h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (BRK _ON) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Brake release request command (BRK_OFF: 22h)

Byte	Command	Response
1	BRK_OFF (22h)	BRK_OFF (22h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	1 ms or less
Subcommand expansion	Refer to p.50.
Description	• Turn ON the MBC output. Note that this command is valid when memory switch 4 (brake operation setting) is set to 0. MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If the following condition applies, a warning will generate and the command will be ignored: · When bit 9 of memory switch 4 is set to 1 (02h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (BRK _OFF) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Sensor ON command (SENS_ON: 23h)

Byte	Command	Response
1	SENS_ON (23h)	SENS_ON (23h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	Return response data. MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SENS_ON) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Sensor OFF command (SENS_OFF: 24h)

Byte	Command	Response
1	SENS_OFF (24h)	SENS_OFF (24h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	Return response data. MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SENS_OFF) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Motion stop request command (HOLD: 25h)

Byte	Command	Response
1	HOLD (25h)	HOLD (25h)
2	0	ALARM
3	OPTION	STATUS
4		
5	HOLD_MOD	MONITOR1
6 to 8	0	
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

<table>
<tr><td>Processing category</td><td>Motion command</td></tr>
<tr><td>Synchronization category</td><td>Asynchronous command</td></tr>
<tr><td>Processing time</td><td>Within the communication period</td></tr>
<tr><td>Subcommand expansion</td><td>Refer to p.50.</td></tr>
<tr><td>Description</td><td>• Decelerate the motor at the specified deceleration until it stops. Latch processing by LATCH and EX_POSING is cancelled.
HOLD_MOD: see table below.
MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian)
<table><tr><th>HOLD_MOD</th><th>Description</th></tr><tr><td>0</td><td>Deceleration stop</td></tr><tr><td>1</td><td>Immediate stop</td></tr></table>• If the following condition applies, a warning will generate and the command will be ignored:
· When the OPTION or HOLD_MOD is outside the specified range (03h).</td></tr>
<tr><td>Confirmation of completion (Process performed by the master)</td><td>• Completion is confirmed based on byte 1 (HOLD) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.
• Whether the motor has stopped completely is confirmed based on bit 8 (DEN=1) of the status field (p.51).</td></tr>
</table>

■ Servo status monitor command (SMON: 30h)

Byte	Command	Response
1	SMON (30h)	SMON (30h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Data communication command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	Monitors the driver status. SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SMON) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Servo ON command (SV_ON: 31h)

Byte	Command	Response
1	SV_ON (31h)	SV_ON (31h)
2	0	ALARM
3	OPTION	STATUS
4		
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	60 ms or less
Subcommand expansion	Refer to p.50.
Description	• Excite the motor. SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When an alarm that cuts off the motor current is present (02h) · When the main power supply is cut off (02h).
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SV _ON) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Servo OFF command (SV_OFF: 32h)

Byte	Command	Response
1	SV_ON (32h)	SV_ON (32h)
2	0	ALARM
3	0	STATUS
4	0	
5 to 8	0	MONITOR1
9 to 12	0	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Control command
Synchronization category	Asynchronous command
Processing time	The MBC output will turn OFF within 1 ms, but motor excitation will not be turned off until approx. 200 ms thereafter.
Subcommand expansion	Refer to p.50.
Description	Turn off motor excitation. SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (SV _OFF) and bit 2 (CMDRDY=1) of the status field (p.51), in the response.

■ Interpolation feed (INTERPOLATE :34h)

Byte	Command	Response
1	INTERPOLATE (34h)	INTERPOLATE (34h)
2	0	ALARM
3	OPTION	STATUS
4		
5 to 8	TPOS	MONITOR1
9 to 12	VFF (Invalid)	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Synchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform interpolation feed. The interpolation position for each communication period is specified as an absolute position. TPOS: Interpolation position (command position, little endian) −2,147,483,648≦TPOS≦+2,147,483,647 (Absolute position: −8,388,608≦TPOS≦ +8,388,607) VFF: Speed feed-forward command value (command unit/s) This item is not supported and the data in this field, if set, will be ignored. SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) · When the difference from the previous TPOS value exceeds the lower or higher limit (−4096 to +4096) (03h)
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (INTERPOLATE) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of position is confirmed based on bit 8 (DEN=1) of the status field (p.51).

■ Positioning (POSING: 35h)

Byte	Command	Response
1	POSING (35h)	POSING (35h)
2	0	ALARM
3 4	OPTION	STATUS
5 to 8	TPOS	MONITOR1
9 to 12	TSPD	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform positioning to the target position (TPOS) at the target speed (TSPD). The target speed or target position can be changed in the middle of positioning. TPOS: Target speed (command position, little endian) −2,147,483,648≦TPOS≦+2,147,483,647 (Absolute position: −8,388,608≦TPOS≦+8,388,607) TSPD: Target speed (command unit/s, little endian) 0≦TSPD≦500,000 (Push-motion operation: 0≦TSPD≦5000) SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) · When the target speed (TSPD) is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (POSING) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of positioning is confirmed based on bit 8 (DEN=1) of the status field (p.51).

■ Constant-speed feed (FEED: 36h)

Byte	Command	Response
1	FEED (36h)	FEED (36h)
2	0	ALARM
3 4	OPTION	STATUS
5 to 8	0	MONITOR1
9 to 12	TSPD	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform constant-speed feed at the target speed (TSPD). The motion stop request command (HOLD: 25h) is used to stop the feed. The target speed can be changed in the middle of positioning. TSPD: Target speed (command unit/s, little endian) −500,000≦TSPD≦+500,000 SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) · When the target speed (TSPD) is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (FEED) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of position-command output is confirmed based on bit 8 (DEN=1) of the status field (p.51).

■ Interpolation feed with position latch function (LATCH: 38h)

Byte	Command	Response
1	LATCH (38h)	LATCH (38h)
2	LT_SGNL	ALARM
3	OPTION	STATUS
4		
5 to 8	TPOS	MONITOR1
9 to 12	VFF (Disable)	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform interpolation feed and latch the current position data with an external latch signal. TPOS: Interpolation position (command position, little endian) −2,147,483,648≦TPOS≦+2,147,483,647 (Absolute position: −8,388,608≦TPOS≦ +8,388,607) VFF: Speed feed-forward command value (command unit /s) This item is not supported and the data in this field, if set, will be ignored. SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) Upon input of an external latch signal, the current position will be output to the latch position (LPOS). In addition, the current position will be output to MONITOR2 forcibly during one communication period. (Furthermore, the data in SEL_MON2 will be forcibly changed to LPOS.) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) · When the difference from the previous TPOS value exceeds the lower or higher limit (−4096 to +4096) (03h)
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (LATCH) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of position is confirmed based on bit 8 (DEN=1) of the status field (p.51).

■ External input positioning (EX_POSING: 39h)

Byte	Command	Response
1	EX_POSING (39h)	EX_POSING (39h)
2	LT_SGNL	ALARM
3	OPTION	STATUS
4		
5 to 8	TPOS	MONITOR1
9 to 12	TSPD	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform positioning to the target position (TPOS) at the target speed (TSPD). If an external latch signal is input in the middle of positioning, the motor will move by the final travel distance for external positioning, from the current position, and then decelerate to a stop. The target position can be changed in the middle of positioning until an external latch signal is input. TPOS: Target position (command position, little endian) −2,147,483,648≦TPOS≦+2,147,483,647 (Absolute position: −8,388,608≦TPOS≦ +8,388,607) TSPD: Target speed (command unit /s、little endian) 0≦TSPD≦500,000 SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) Upon input of an external latch signal, the current position will be output to the latch position (LPOS). In addition, the current position will be output to MONITOR2 forcibly during one communication period. • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) ・When the target speed (TSPD) is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (EX_POSING) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of position-command output is confirmed based on bit 8 (DEN=1) of the status field (p.51).

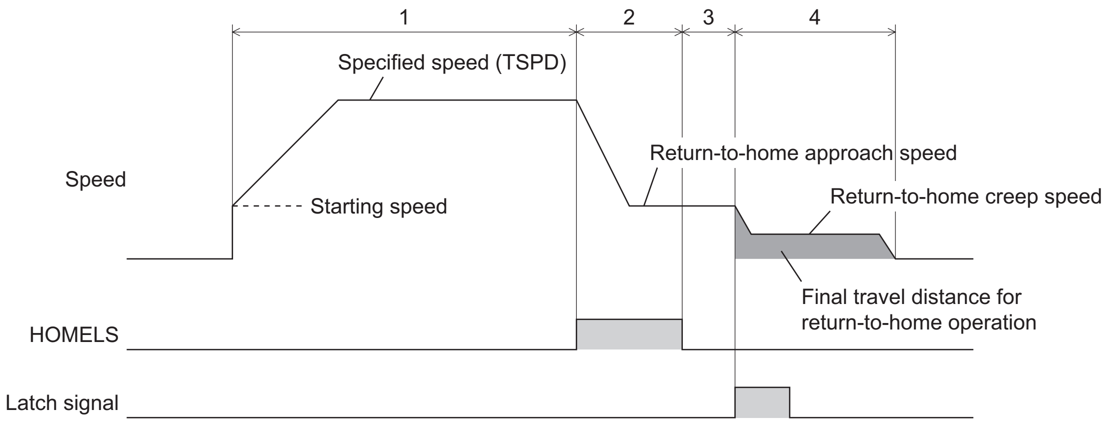
■ Return-to-home (ZRET: 3Ah)

Byte	Command	Response
1	ZRET (3Ah)	ZRET (3Ah)
2	LT_SGNL	ALARM
3 4	OPTION	STATUS
5 to 8	0	MONITOR1
9 to 12	TSPD	MONITOR2
13	SEL_MON1/2	SEL_MON1/2
14	0	IO_MON
15	0	
16	WDT	RWDT
17 to 31	Subcommand area	Subcommand area

Processing category	Motion command
Synchronization category	Asynchronous command
Processing time	Within the communication period
Subcommand expansion	Refer to p.50.
Description	• Perform return-to-home operation using the HOMELS signal and an external latch signal. The target speed cannot be changed in the middle of return-to-home operation. TSPD: Target speed (command unit/s, little endian) 0≦TSPD≦500,000 SEL_MON1/2: Select monitor MONITOR1/2: Monitor information selected by SEL_MON1/2 (little endian) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the servo is off (02h) · When the motor is decelerating to a stop (02h) · When the target speed (TSPD) is outside the specified range (03h).
Confirmation of completion (Process performed by the master)	• Completion is confirmed based on byte 1 (ZRET) and bit 2 (CMDRDY=1) of the status field (p.51), in the response. • Completion of position-command output is confirmed based on bit 6 (ZPOINT=1) and bit 8 (DEN=1) of the status field (p.51).

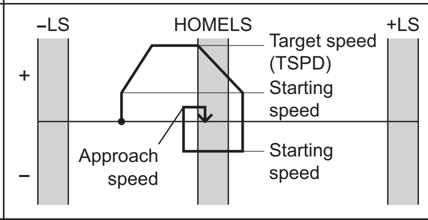
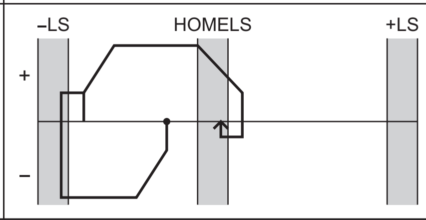
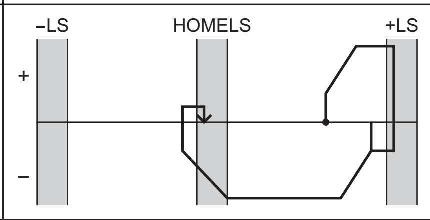
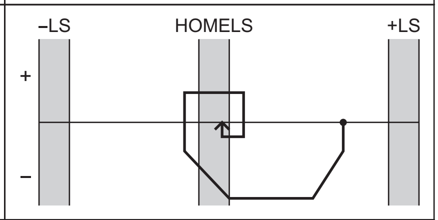
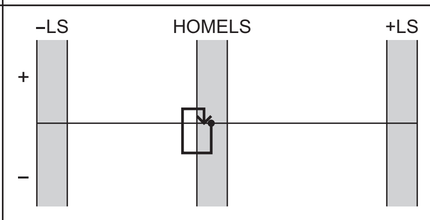
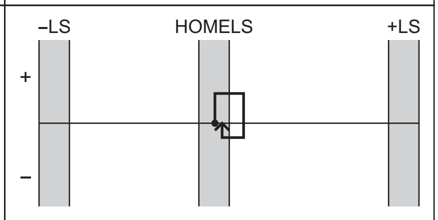
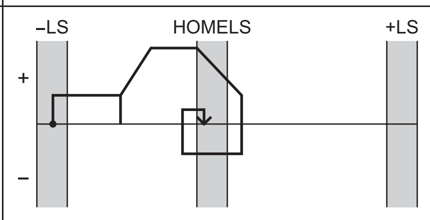
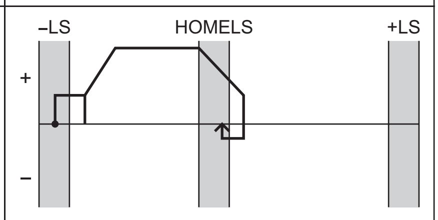
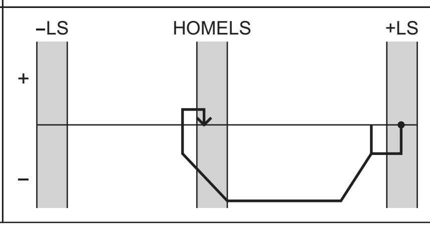
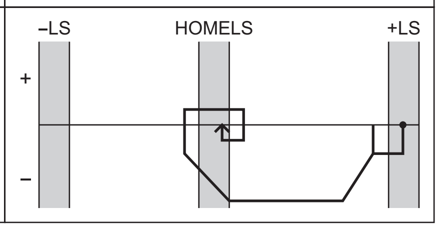
Note Set the return-to-home operation using the return-to-home operation selection parameter.

- Example of return-to-home operation selection=0 (return-to-home operation type 1)



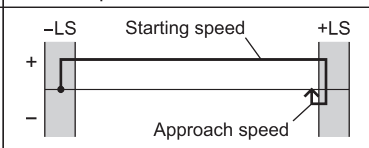
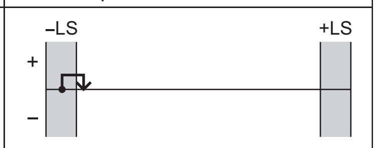
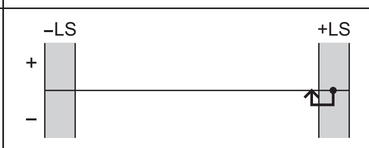
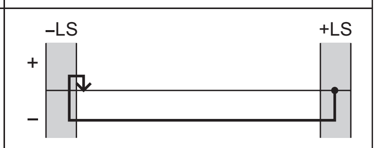
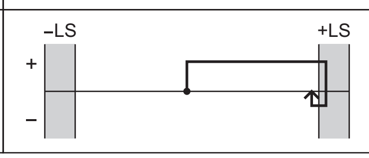
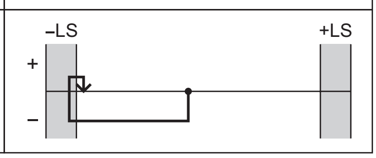
1. The motor feeds in the return-to-home direction specified by the memory switch 4 parameter, at the target speed (TSPD) specified by the ZRET command.
2. When the HOMELS turns ON, the motor decelerates to the home approach speed*.
3. When the HOMELS turns OFF, the motor waits for input of an external latch signal.
4. Upon input of an external latch signal, the motor decelerates to the return-to-home creep speed and moves to a position which is away from the current position by the final travel distance for return-to-home operation*, and then stops.
   If the motor cannot decelerate to a stop within the final travel distance for return-to-home operation, or if the specified final travel distance for return-to-home operation is a negative value, the motor decelerates to a stop and then moves to a position corresponding to the final travel distance for return-to-home operation.
   The return-to-home creep speed is set by a parameter.
5. Once the positioning is complete, the current position is set as 0.

- Example of return-to-home operation selection=1 (return-to-home operation type 2: 3-sensor mode)

Starting position of return-to-home operation	Return-to-home operation direction: +	Return-to-home operation direction: –
Between HOMELS and –LS		
Between HOMELS and +LS		
HOMELS		
–LS		
+LS		

Home offset operation is not performed.

- Example of return-to-home operation selection=1 (return-to-home operation type 2: 2-sensor mode)

Starting position of return-to-home operation	Return-to-home operation direction: +	Return-to-home operation direction: –
–LS		
+LS		
Between –LS and +LS		

Home offset operation is not performed.

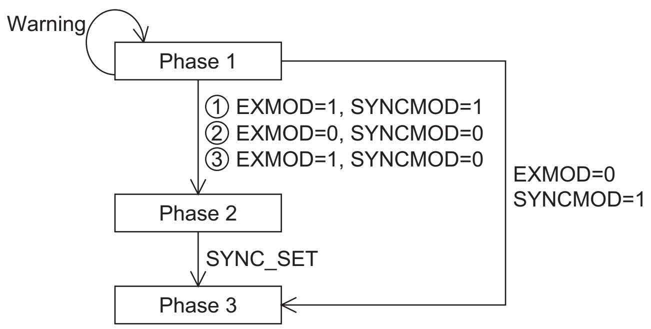
■ MECHATROLINK-I connection request command (CONNECT: 0Eh)

Byte	Command	Response
1	CONNECT (0Eh)	CONNECT (0Eh)
2	0	ALARM
3	0	STATUS
4	0	
5	VER	VER
6	COM_MODE	COM_MODE
7	COM_TIME	COM_TIME
8 to 15	0	0
16	WDT	RWDT

Processing category	Network command
Synchronization category	Asynchronous command
Processing time	More than the communication period
Subcommand expansion	Disable
Description	• Request MECHATROLINK connection to be established. Upon completion of this command, the system will switch to phase 2 or 3 and communication with the master station will be enabled. Set the communication mode and communication period. VER: Set the version VER=10H (Ver.1.0) COM_MODE: Communication mode (see table below.) COM_TIME: Communication period (fixed to 2 ms) • If one of the following conditions applies, a warning will generate and the command will be ignored: · When the COM_MODE is outside the specified range (03h). · When the COM_TIME is outside the specified range (03h). · When the MECHATROLINK communication mode is active and VER is not 10H (03h)
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 1 (CONNECT), bit 2 (CMDRDY=1) of the status field (p.51), and set data (VER, COM_MODE, COM_TIME), in the response.

• COM_MODE list

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
SUBCMD	0	0	0	DTMOD		SYNCMOD	EXMOD

Name	Description	Setting
EXMOD	Establishment of connection	0: Standard connection 1: Expanded connection
SYNCMOD	Synchronization setting	0: Perform asynchronous communication (switch to phase 2). Watchdog check is disabled. Synchronous communication commands cannot be used. 1: Perform synchronous communication (switch to phase 3). Watchdog check is enabled. Synchronous communication commands can be used. 
DTMOD	Communication method	00: Single transmission communication
SUBCMD	Subcommand setting	0: Subcommand disabled

Note If EXMOD=1, the system switches to phase 2. The system will switch to 3 when the SYNC_SET command is completed.

9.3 Sub command specifications

MECHATROLINK-II subcommands are specified by the CONNECT command.
These subcommands use bytes 17 to 31 of command/response data.

■ Invalid command (NOP: 00h)

Byte	Command	Response
17	NOP (00h)	NOP (00h)
18	0	SUBSTATUS
19 to 31	0	0

Processing category	Network command
Processing time	Within the communication period
Description	Invalid command or no command. The current status (substatus) is returned from the slave as a response.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (NOP), and bit 2 (SBCMDRDY=1) of the substatus field (p.54), in the response.

■ Parameter read command (PRM_RD: 01h)

Byte	Command	Response
17	PRM_RD (01h)	PRM_RD (01h)
18	0	SUBSTATUS
19	No.	No.
20		
21	SIZE	SIZE
22 to 31	0	PARAMETER

Processing category	Data communication command
Processing time	Within the communication period
Description	Specify the parameter number and data size and read the parameter. The function is the same as that of the main command PRM_RD.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (PRM_RD), bit 2 (SBCMDRDY=1) of the substatus field (p.54), No. and SIZE, in the response.

■ Parameter write command (PRM_WR: 02h)

Byte	Command	Response
17	PRM_WR (02h)	PRM_WR (02h)
18	0	SUBSTATUS
19	No.	No.
20		
21	SIZE	SIZE
22 to 31	PARAMETER	PARAMETER

Processing category	Data communication command
Processing time	150 ms or less
Description	Specify the parameter number and data size and write the parameter to the RAM. The function is the same as that of the main command PRM_WR.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (PRM_WR), bit 2 (SBCMDRDY=1) of the substatus field (p.54), No., SIZE and PARMETER, in the response.

■ Alarm/warning read command (ALM_RD: 05h)

Byte	Command	Response
17	ALM_RD (05h)	ALM_RD (05h)
18	0	SUBSTATUS
19	ALM_RD_MODE	ALM_RD_MODE
20 to 29	0	ALM_DATA (new) ALM_DATA (old)
30		0
31		0

Processing category	Data communication command
Processing time	See table below.
Description	Read each present alarm/warning using an alarm code. The function is the same as that of the main command ALM_RD.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (ALM _RD) and bit 2 (SBCMDRDY=1) of the substatus field (p.54), in the response.

- ALM_RD_MODE list

ALM_RD_MODE	Description	Processing time
0	Read the alarm and warning currently present. (Only byte 20 is used.)	Within the communication period
1	Read alarm records (up to 10).	150 ms or less

■ Nonvolatile parameter write command (PPRM_WR: 1Ch)

Byte	Command	Response
17	PPRM_WR (1Ch)	PPRM_WR (1Ch)
18	0	SUBSTATUS
19	No.	No.
20		
21	SIZE	SIZE
22 to 31	PARAMETER	PARAMETER

Processing category	Data communication command
Processing time	150 ms or less
Description	Specify the parameter number, data size and parameter data and write the data to the NV memory and RAM. The function is the same as that of the main command PPRM_WR.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (PPRM_WR), bit 2 (SBCMDRDY=1) of the substatus field (p.54), No., SIZE and PARMETER, in the response.

Note The NV memory can be rewritten approx. 100,000 times.

■ Servo status monitor command (SMON: 30h)

Byte	Command	Response
17	SMON (30h)	SMON (30h)
18	0	SUBSTATUS
19	SEL_MON3/4	SEL_MON3/4
20 to 23	0	MONITOR3
24 to 27	0	MONITOR4
28 to 31	0	0

Processing category	Data communication command
Processing time	Within the communication period
Description	Monitors the driver status. SEL_MON3/4: Select monitor The function is the same as that of the main command SMON.
Confirmation of completion (Process performed by the master)	Completion is confirmed based on byte 17 (SMON) and bit 2 (SBCMDRDY=1) of the substatus field (p.54), in the response.

■ Combination of MECHATROLINK-Ⅱ main commands and subcommands

In MECHATROLINK-II communication, main commands and subcommands can be confirmed according to the table below.

Code	Main command	Sub command					
		NOP (00h)	PRM_RD (01h)	PRM_WR (02h)	ALM_RD (05h)	PPRM_WR (1Ch)	SMON (30h)
00h	NOP	○	○	○	○	○	○
01h	PRM_RD	○	×	×	×	×	○
02h	PRM_WR	○	×	×	×	×	○
03h	ID_RD	○	○	○	○	○	○
04h	CONFIG	○	×	×	×	×	○
05h	ALM_RD	○	×	×	×	×	○
06h	ALM_CLR	○	×	×	×	×	○
0Dh	SYNC_SET	○	×	×	×	×	○
0Eh	CONNECT	○	×	×	×	×	×
0Fh	DISCONNECT	○	×	×	×	×	×
1Ch	PPRM_WR	○	×	×	×	×	○
20h	POS_SET	○	×	×	×	×	○
21h	BRK_ON	○	×	×	×	×	○
22h	BRK_OFF	○	×	×	×	×	○
23h	SENS_ON	○	×	×	×	×	○
24h	SENS_OFF	○	×	×	×	×	○
25h	HOLD	○	○	○	○	○	○
30h	SMON	○	○	○	○	○	○
31h	SV_ON	○	○	○	○	○	○
32h	SV_OFF	○	○	○	○	○	○
34h	INTERPOLATE	○	○	○	○	○	○
35h	POSING	○	○	○	○	○	○
36h	FEED	○	○	○	○	○	○
38h	LATCH	○	○	○	○	○	○
39h	EX_POSING	○	○	○	○	○	○
3Ah	ZRET	○	○	○	○	○	○

○: Can be combined.
×: Cannot be combined.

9.4 Command data field

This section explains the data of main commands and subcommands.

■ Option (OPTION) field

Append a motion command function.

Reserved area in command	byte 3 to byte 4
Reserved area in response	None
Supported commands *	SV_ON, HOLD, INTERPOLATE, POSING, FEED, LATCH, EX_POSING, ZRET

* When data relating to push-motion operation is set (PUSH_EN, PUSH_CURR), POSING is the only command supported by this field. PUSH_EN and PUSH_CURR can no longer be changed once push-motion operation is started. All other commands will be ignored.

- Bit assignments

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
PUSH_EN	0	0	ACCFIL		0	0	0

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
PUSH_CURR							

- Description of the data

Name	Description	Setting
ACCFIL	acceleration/deceleration filter selection	0: Linear acceleration/deceleration
PUSH_EN	Push-motion operation selection	0: The push-motion operation become ineffective. 1: The push-motion operation become effective.
PUSH_CURR	Push-motion operation current	0 to 50 [%] (0 to 32Hex): Sets the push-motion operation current as a percentage of the rated current.

■ Latch signal selection (LT_SGNL) field

Select a signal used to latch the position data.

Reserved area in command	byte 2
Reserved area in response	None
Supported commands	LATCH, EX_POSING, ZRET

- Bit assignments

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	0	0	0	0	0	LT_SGNL	

- Description of the data

Name	Description	Setting
LT_SGNL	Latch signal selection	0: TIM signal (Refer to p.66 for details.) 1: EXT1 (external latch signal 1) pin No.10 of user I/O 2: EXT2 (external latch signal 2) pin No.11 of user I/O 3: EXT3 (external latch signal 3) pin No.12 of user I/O

■ Status (STATUS) field

Monitors the driver status.

Reserved area in command	None
Reserved area in response	byte 3 to byte 4
Supported commands	All main command

- Bit assignments

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
PSET	ZPOINT	0	PON	SVON	CMDRDY	WARNG	ALM

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
0	MOVE	N_SOT	P_SOT	NEAR	L_CMP	0	DEN

- Description of the data

Name	Description	Setting	Explanation
ALM	Alarm	0: Alarm not present 1: Alarm present	0 is set if the alarm code is 00h. 1 is set if an alarm (whose alarm code is other than 00h to 03h) is present.
WARNG	Warning	0: Warning not present 1: Warning present	0 is set if no MECHATROLINK communication error is present. 1 is set if a MECHATROLINK communication error (whose alarm code is 01h to 03h) is present.
CMDRDY	Command ready	0: Command not acceptable (Busy) 1: Command acceptable (Ready)	0 is set if the next command cannot be executed because the current command is still being executed and cannot be completed within the communication period. 1 is set if the next command can be executed.
SVON	Servo ON	0: Servo OFF 1: Servo ON	0 is set if excitation is off (the servo is off). 1 is set if excitation is on (the servo is on).
PON	Main power supply	0: Main power supply OFF 1: Main power supply ON	0 is set if the main power supply is off. 1 is set if the main power supply is on.
ZPOINT	Home position	0: Home position out of range 1: Home position in range	0 is set if the command position is out of the home position range set by a parameter. 1 is set if the command position is within the home position range set by a parameter. For Home position range parameter, refer to p.58.
PSET	Positioning	0: Positioning complete out of range 1: Positioning complete in range	0 is set if the positioning is not yet complete. 1 is set if the positioning has already completed. * A positioning is considered complete when pulse output has completed (DEN=1) and the rotor has stabilized at a position within ±1.8° of the command position. If the speed filter time constant is large, a response on completion of positioning may be delayed.
DEN	Pulse output	0: Pulse output in progress 1: Pulse output complete	0 is set if output of command pulses is still in progress. 1 is set if output of command pulses has completed. * Completion of output is confirmed based on the command effective before filtering is applied by the speed filter time constant.
L_CMP	Latch	0: Latch not yet complete 1: Latch complete	0 is set if the system is yet to switch to latch signal input operation. 1 is set if the LATCH, EX_POSING or ZRET command is being executed or the system has already switched to latch signal input operation. * The setting remains 0 if the applicable command is other than LATCH, EX_POSING or ZRET. Even when L_CMP=1, 0 will be set once the next motion command is executed.
NEAR	Positioning near	0: Positioning near out of range 1: Positioning near in range	0 is set if the command position is out of the positioning near band set by a parameter. 1 is set if the command position is within the positioning near band set by a parameter. For positioning near (NEAR) band parameter, refer to p.58.
P_SOT	+Soft limit	0: Positive soft limit in range 1: Positive soft limit out of range	0 is set if the command position is inside of the positive soft limit set by a parameter (the soft limit signal is not detected). 1 is set if the command position is outside of the positive soft limit set by a parameter (the soft limit signal is detected). For soft limit parameter, refer to p.59.
N_SOT	−Soft limit	0: Negative soft limit in range 1: Negative soft limit out of range	0 is set if the command position is inside of the negative soft limit set by a parameter (the soft limit signal is not detected). 1 is set if the command position is outside of the negative soft limit set by a parameter (the soft limit signal is detected).For soft limit parameter, refer to p.59.
MOVE	Operation command	0: No operation command 1: Operation command being executed	0 is set if an operation command is not being executed (an internal oscillation function of the driver is not active). 1 is set if an operation command is being executed (an internal oscillation function of the driver is active).

■ Monitor selection (SEL_MON1～4) field

Select the monitor code.

Reserved area in command	Main command: byte 13 (SEL_MON1/2) Sub command: byte 19 (SEL_MON3/4)
Reserved area in response	Main command: byte 13 (SEL_MON1/2) Sub command: byte 19 (SEL_MON3/4)
Supported commands	SV_ON, SV_OFF, HOLD, INTERPOLATE, POSING, FEED, LATCH, EX_POSING, ZRET, SMON, SENS_ON, SENS_OFF, BRK_ON, BRK_OFF

- Bit assignments

Set an applicable monitor code if selection is made using SEL_MON1 to 4.

Main command

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
SEL_MON2				SEL_MON1			

Sub command

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
SEL_MON4				SEL_MON3			

- Monitor code

MONITOR will return 0 if a reserved area is specified.

Monitor code	Notation	Description	Unite	Remark
0	POS	Command position	Command unit	
1	MPOS	Command position		Same as POS.
2	PERR	Position deviation		
3	APOS	Feedback position		
4	LPOS	Latch position		
5	IPOS	Command position		Same as POS.
6	TPOS	Positioning target position		
7	Reserved	−	−	
8	FSPD	Feedback speed	Command unit/s	Refreshed every 500 ms (average of travel over 500 ms).
9	CSPD	Command speed		
A	TSPD	Target speed		
B	XCRNT	Excitation current	%	A percentage of the rated current
C	Reserved	−	−	
D	Reserved	−	−	
E	Reserved	−	−	
F	Reserved	−	−	

■ Monitor information (MONITOR 1 to 4) field

Monitor the content of the monitor code selected in the monitor selection field.

Reserved area in command	None
Reserved area in response	Main command: byte 5 to byte 8(MONITOR1) byte 9 to byte 12 (MONITOR2) Sub command: byte 20 to byte 23 (MONITOR3) byte 24 to byte 27 (MONITOR4)
Supported commands	SV_ON, SV_OFF, HOLD, INTERPOLATE, POSING, FEED, LATCH, EX_POSING, ZRET, SMON, SENS_ON, SENS_OFF, BRK_ON, BRK_OFF

■ I/O monitor (IO_MON) field

Monitor the I/O signal statuses of the driver. The ON/OFF status is indicated according to the active logic.

Reserved area in command	None
Reserved area in response	byte 14 to byte 15
Supported commands	SV_ON, SV_OFF, HOLD, INTERPOLATE, POSING, FEED, LATCH, EX_POSING, ZRET, SMON, SENS_ON, SENS_OFF, BRK_ON, BRK_OFF

- Bit assignments

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
EXT2	EXT1	TIM	BSG	ASG	HOMELS	−LS	+LS

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
0	0	0	0	0	ALM	MBC	EXT3

- Description of the data

Name	Description	Setting
+LS	+limit sensor	0: OFF 1: ON
−LS	−limit sensor	
HOMELS	HOME sensor	
ASG	A-phase output	
BSG	B-phase output	
TIM	TIM signal output	
EXT1	External latch signal 1	
EXT2	External latch signal 2	
EXT3	External latch signal 3	
MBC	Electromagnetic brake control output	0: Release the electromagnetic brake 1: Hold the electromagnetic brake
ALM	Alarm output	0: Alarm not present 1: Alarm present

■ Sub status (SUBSTATUS) field

Monitors the sub command status.

Reserved area in command	None
Reserved area in response	byte 18
Supported commands	All sub commands

- Bit assignments

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	0	0	0	0	SBCMDRDY	SBWARNG	SBALARM

- Description of the data

Name	Description	Setting	
SBALARM	Sub command alarm	0: Alarm not present 1: Alarm present	0 is set if a communication alarm is not present in response to the subcommand. 1 is set if a communication alarm is present in response to the subcommand.
SBWARNG	Sub command warning	0: Warning not present 1: Warning present	0 is set if a MECHATROLINK communication error is not present in response to the subcommand. 1 is set if a MECHATROLINK communication error (whose alarm code is 01h to 03h) is present in response to the subcommand.
SBCMDRDY	Sub command ready (Subcommand acceptable)	0: Subcommand not acceptable (Busy) 1: Subcommand acceptable (Ready)	0 is set if the next subcommand cannot be executed because the current subcommand is still being executed and cannot be completed within the communication period. 1 is set if the next subcommand can be executed.

9.5 Parameter list

The parameters that can be set in the **ARL** series MECHATROLINK-Ⅱ driver are listed below.
You can use the **MEXE02** to set parameters.
Supported commands: PRM_RD, PRM_WR, PPRM_WR

No.	Name	Size (Byte)	Read/ Write	Setting unit	Setting range	Initial value
0000h*1	Parameter switch	2	R	−	8000h	8000h (fixed)
0001h	Memory switch 1	2	R/W	−	0000h to FFFFh	0
0002h	Memory switch 2	2	R/W	−	0000h to FFFFh	0
0003h	Memory switch 3	2	R/W	−	0000h to FFFFh	0
0004h	Memory switch 4	2	R/W	−	0000h to FFFFh	0
0005h	Electronic gear (numerator)	2	R/W	−	1 to 100	1
0006h	Electronic gear (denominator)	2	R/W	−	1 to 100	1
0007h*1	Basic resolution	2	R	p/r	10000	10000 (fixed)
0008h	Home position range	2	R/W	Command unit*3	0 to 65535	0
0009h*1	Home position offset (absolute specification)	4	R	Command unit*3	Not supported	0
000Ah*1	Positioning complete output band	2	R	360/51200°	250 (=±1.8°)	256 (fixed)
000Bh	Positioning near (NEAR) band	2	R/W	Command unit*3	0 to 65535	0
000Ch	+Soft limit	4	R/W	Command unit*3	−8,388,608 to +8,388,607	+8,388,607
000Dh	−Soft limit	4	R/W	Command unit*3	−8,388,608 to +8,388,607	−8,388,608
000Eh	Starting speed	2	R/W	100 command unit/s*3	0 to 5000	10
000Fh	Linear acceleration constant (acceleration rate)	2	R/W	10000 command unit/s^2 *3	1 to 10000	100
0010h	Linear deceleration constant (deceleration rate)	2	R/W	10000 command unit/s^2 *3	1 to 10000	100
0011h	Final travel distance for external positioning	4	R/W	Command unit*3	−8,388,608 to +8,388,607	0
0012h	Return-to-home approach speed	2	R/W	100 command unit/s*3	0 to 5000	10
0013h	Return-to-home creep speed	2	R/W	100 command unit/s*3	0 to 5000	10
0014h	Final travel distance for return-to-home operation	4	R/W	Command unit*3	−8,388,608 to +8,388,607	0
0015h	Operating current	2	R/W	%	0 to 100	100
0016h	Standstill current	2	R/W	%	0 to 50	50
1001h*2	Speed filter time constant	2	R/W	ms	1 to 100	3
1002h*2	Overload detection time	2	R/W	0.1 s	1 to 250	50
1003h*2	Amount of overflow rotation	2	R/W	0.02 rotation	1 to 32000	150
1004h*2	Return-to-home operation selection	2	R/W	−	0000h to FFFFh	0
1005h*2	Resetting the absolute position loss error alarm	2	W	−	0: Not reset 1: Reset the alarm and confirm the coordinates.	0
1006h*2	Overtravel mask	2	R/W	−	0: Output the ±LS signal detection alarm. 1: Do not output the ±LS signal detection alarm.	0

*1 This parameter is fixed. The setting cannot be changed.
*2 This is a vendor parameter and fixed.
*3 Command unit: One unit of travel distance according to the motor resolution set by the electronic gear ratio

■ Parameter switch

The **ARL** series MECHATROLINK-Ⅱ driver uses the standard parameters as is, but it also offers unique parameters that can be set as an option. Accordingly, the parameter switch is always set to 8000h (read-only).

Bit	Name	Setting
0	Electronic gear (numerator, denominator) Defined separately.	0: Standard parameters are used. (initial value) 1: Unique parameters are used.
1	Basic resolution Defined separately.	
2	Home position range Defined separately.	
3	Home position offset Defined separately.	
4	Positioning complete output band Defined separately.	
5	Positioning near (NEAR) band Defined separately.	
6	Soft limit (forward, reverse) Defined separately.	
7	Starting speed Defined separately.	
8	Acceleration/deceleration rate Defined separately.	
9	Final travel distance for external positioning Defined separately.	
10	Return-to-home operation speed (approach speed, creep speed) Defined separately.	
11	Final travel distance for return-to-home operation Defined separately.	
12	Operating current Defined separately.	
13	Standstill current Defined separately.	
14	Reserved	−
15	Fixed parameters other than the standard parameters are used.	0: Standard parameters are only used. 1: Unique parameters are also used. (initial value)

■ Memory switch 1

Bit	Name	Setting	Description	Initial value
0	−	0: Fixed	−	−
1	−	0: Fixed	−	−
2	+LS mask	0: Input of the +LS signal causes the motor to stop according to the stop action specified by bit 8. 1: Input of the +LS signal does not cause the motor to stop.	Enable/disable the +LS input. Even though this bit is set to 1, the +LS input will be enabled automatically if the return-to-home operation type is 2, but only when the ZRET command is executed.	0
3	−LS mask	0: Input of the −LS signal causes the motor to stop according to the stop action specified by bit 8. 1: Input of the −LS signal does not cause the motor to stop.	Enable/disable the −LS input. Even though this bit is set to 1, the −LS input will be enabled automatically if the return-to-home operation type is 2, but only when the ZRET command is executed.	0
4 to 7	−	0: Fixed	−	−
8	Overtravel stop action	0: The motor decelerates to a stop when the ±LS is input. 1: The motor stop immediately when the ±LS is input.	Sets how the motor should stop when the overtravel is detected.	0
9 to 13	−	0: Fixed	−	−
14	Absolute function	0: Disable (incremental specification) 1: Enable (absolute specification)	Enable/disable the absolute function. Absolute function will become effective only after the power is cycled. For the absolute specification to be used, the optional battery (sold separately) will be required.	0
15	−	0: Fixed	−	−

■ Memory switch 2

Bit	Name	Setting	Description	Initial value
0	Motor rotation direction switching	0:+direction=Rotate to CCW 1: +direction=Rotate to CW	Sets the rotation direction of motor output shaft. The rotation direction is as viewed from the side of the motor's output shaft.	0
1 to 5	−	0: Fixed	−	−
6	Availability of +soft limit check*	0: Do not perform soft limit check on the positive side (disable the +soft limit). 1: Perform soft limit check on the positive side (enable the +soft limit).	Enable/disable soft limit check on the positive side.	0
7	Availability of −soft limit check*	0: Do not perform soft limit check on the negative side (disable the −soft limit). 1: Perform soft limit check on the negative side (enable the −soft limit).	Enable/disable soft limit check on the negative side.	0
8 to 15	−	0: Fixed	−	−

* If soft limit check is enabled, the check will be performed based on the following conditions. In any other condition, the soft limit signal will not be detected even though the soft limit range is exceeded:
· After the ZRET command has been executed
· After the POS_SET command has been issued with REFE=1
· When bit 14 of memory switch 1 (absolute function) is enabled

■ Memory switch 3

Bit	Name	Setting	Description	Initial value
0 to 9	−	0: Fixed	−	−
10	MECHATROLINK communication check mask (for debugging)	0: Perform communication check 1: Do not perform communication check Ignore the command errors (01 to 03)	This function is used for debugging during trial operation, etc. You can forcibly disable the communication alarm check. Normally this bit must be set to 0.	0
11	WDT check mask (for debugging)	0: Perform WDT check 1: Do not perform WDT check Ignore the command error (04)	This function is used for debugging during trial operation, etc. You can forcibly disable the WDT alarm check. Normally this bit must be set to 0.	0
12 to 15	Communication error count	0 to F	If a communication error has occurred consecutively for the number of times set here, the communication error handling process (parallel off, excitation off) is performed. Normally this bit must be set to 0 (system-specific setting: twice).	0

■ Memory switch 4

Bit	Name	Setting	Description	Initial value
0	−	0: Fixed	−	−
1	Return-to-home direction	0: Rotate to +direction. 1: Rotate to −direction.	The motor direction is switched upon receiving of the return-to-home command (ZRET: 3Ah).	0
2 to 8	−	0: Fixed	−	−
9	Brake operation	0: Operate using the BRK_ON/BRK_OFF command 1: Disable the BRK_ON/BRK_OFF command	Enable/disable the MBC signal, which is a user I/O, upon receiving the brake actuation request command (BRK_ON: 21h) or brake release request command (BRK_OFF: 22h).	0
10	+LS signal logic	0: Positive logic (normally closed) 1: Negative logic (normally open)	Set the +LS input logic.	0
11	−LS signal logic	0: Positive logic (normally closed) 1: Negative logic (normally open)	Set the −LS input logic.	0
12	HOMELS signal logic	0: Positive logic (normally closed) 1: Negative logic (normally open)	Set the HOMELS input logic.	0
13 to 15	−	0: Fixed	−	−

■ Electronic gear (numerator), electronic gear (denominator)

The motor runs based on the speed and travel amount calculated by multiplying the respective data by the electronic gear ratio.

$$\text{Motor resolution [p/r]} = 10000 \times \frac{\text{Electronic gear B (numerator)}}{\text{Electronic gear A (denominator)}}$$

$$\text{Motor rotating speed [r/min]} = \frac{1}{10000} \times \frac{\text{Electronic gear (denominator)}}{\text{Electronic gear (numerator)}} \times \text{Operating speed} \times 60$$

∗ Basic resolution setting:10000

Note
- The motor resolution set by the electronic gear ratio should be kept within the following range: 500 [p/r]≦Motor resolution≦10000 [p/r]
- When the electronic gear is changed, the main power supply and control power supply must be cycled.

Setting example) Ball screw

Ball screw lead : 6 mm (0.236 in.)
Minimum travel amount : 0.001 mm (0.00004 in.)
Gear ratio : 1 (No speed reduction mechanism between the motor and ball screw)

$$\text{Motor resolution [p/r]} = 10000 \times \frac{\text{Electronic gear (numerator)}}{\text{Electronic gear (denominator)}} = \frac{\text{Ball screw lead}}{\text{Minimum travel amount}} \times \text{Gear ratio}$$

$$10000 \times \frac{\text{Electronic gear (numerator)}}{\text{Electronic gear (denominator)}} = \frac{6\text{ mm}}{0.001\text{ mm}} \times 1$$

$$\frac{\text{Electronic gear (numerator)}}{\text{Electronic gear (denominator)}} = \frac{6}{10}$$

Accordingly, the denominator and numerator of the electronic gear fraction become 10 and 6, respectively, thus giving a motor resolution of 6000 [p/r].

■ Home position range

Set the Home position detection range (ZPOINT). ZPOINT=1 if the current position is in the home position range, or ZPOINT=0 if the current position is out of the home position range.

■ Positioning near (NEAR) band

Set the positioning near (NEAR) band in NEAR of the STATUS field.
NEAR=1 if the current position is within the positioning near range relative to the target position (TPOS), or NEAR=0 if the current position is outside this range, regardless of completion of output (DEN=1).

■ +Soft limit, −Soft limit

Set the +soft limit and −soft limit
Make sure the +soft limit is greater than the −soft limit.

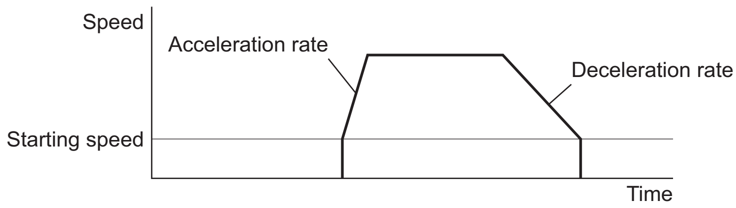
■ Starting speed

Set the starting speed. The starting speed will become effective in motion commands.

■ Linear acceleration constant (acceleration rate), Linear deceleration constant (deceleration rate)

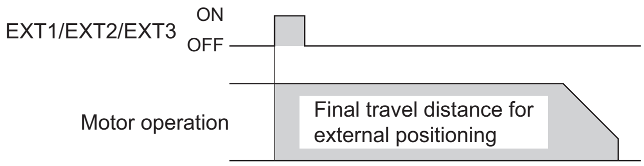
Set the acceleration rate and deceleration rate. These parameters will become effective in motion commands.

Operation example



■ Final travel distance for external positioning

Set the distance to be traveled from the position where an external latch signal (EXT1 to 3) is input when the external input positioning command (EX_POSING) is executed.

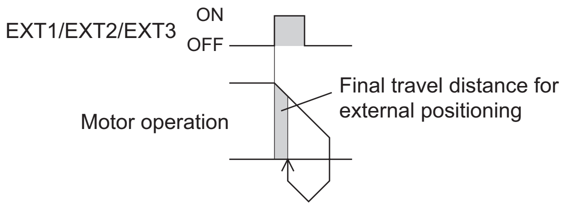


There may be a delay of approx. 1.0 to 1.5 ms following the ON edge of the external latch input signal, before external positioning operation is actually started, due to the effect of the I/O sampling time set in the software. Accordingly, the travel amount from the ON edge of the external latch input signal is calculated as follows:

Travel amount = Distance traveled during the sampling delay time (operating speed × delay time)
+ Final travel distance for external positioning

Also note that the stop position may vary by a maximum of 0.5 ms due to the effect of sampling.

If the opposite direction is specified or the final travel distance for external positioning is short, the motor will decelerate to a stop according to the deceleration pattern (DEN=0) and then pull back to the target position (DEN=1).



■ Return-to-home approach speed

Set the speed to be applied when the HOMELS signal turns ON while the return-to-home command (ZRET) is being executed.

Note If the latch signal selection (LT_SGNL) field is set to “TIM signal,” be sure to set the return-to-home approach speed to 500 Hz or less. If this speed is greater than 500 Hz, the TIM signal may not be detected properly.

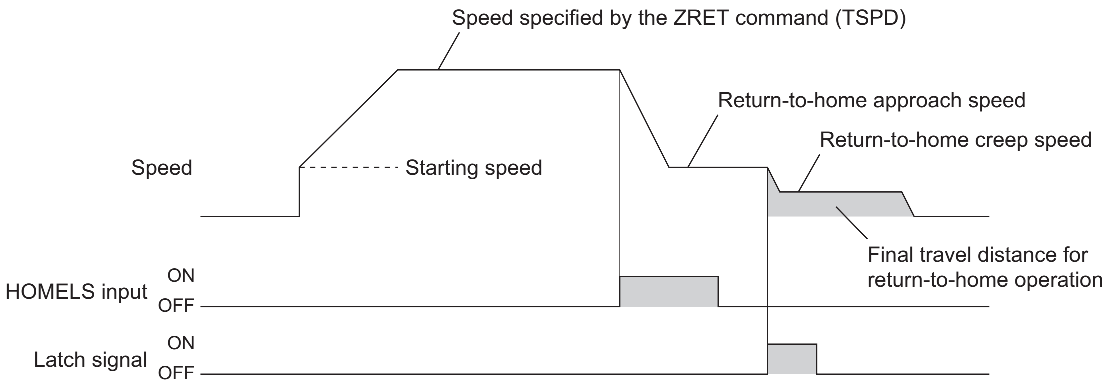
■ Return-to-home creep speed

Set the return-to-home creep speed to be applied when the HOMELS signal switches from ON to OFF while the return-to-home command (ZRET) is being executed.

■ Final travel distance for return-to-home operation

Set the distance to be traveled to the home from the position where a latch signal is input while the return-to-home command (ZRET) is being executed.
If the opposite direction is specified or the final travel distance for external positioning is short, the motor will decelerate to a stop according to the deceleration pattern (DEN=0) and then pull back to the target position (DEN=1).
The latch signal corresponds to the setting in the latch signal selection field.

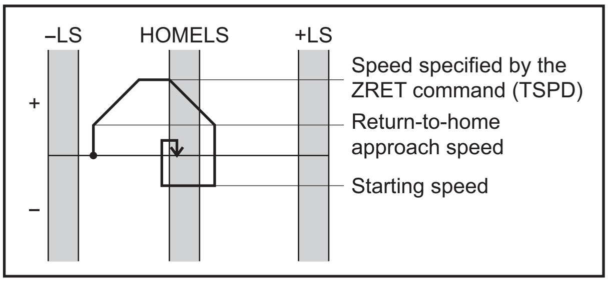
- Return-to-home operation selection=0 (return-to-home operation type 1)



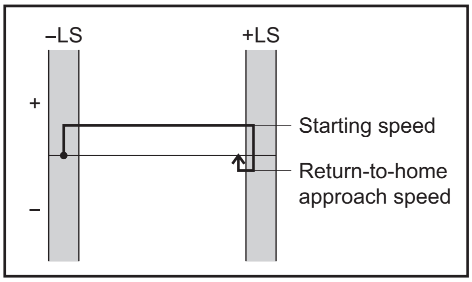
- Return-to-home operation selection=1 (return-to-home operation type 2)

There is no final travel distance for return-to-home operation.

3-sensor mode



2-sensor mode



■ Operating current

Sets the motor operating current as a percentage of the rated current.

■ Standstill current

Sets the motor standstill current as a percentage of the rated current. The operating current will switch to the standstill current in approx. 100 ms after the motor stops.

Note Setting the motor's current slightly lower may have a favorable effect in terms of suppressing increases in motor temperature and reducing vibration, if the load is relatively small and there is extra torque. However, If the standstill current is too low, torque, holding torque and starting characteristics may be dropped. Set an appropriate value.

■ Speed filter time constant

When the speed filter time constant is set, vibration or shock can be suppressed, and starting/stopping of the motor will become smooth.

Note An excessively high filter time constant will result in lower synchronicity with commands. Set an appropriate value according to the specific load and purpose.

■ Overload detection time

Set the time until an overload protection alarm is detected.

■ Amount of overflow rotation

Set the amount of overflow rotation until an excessive position deviation alarm is detected.

■ Return-to-home operation selection

Change the return-to-home operation type. For return-to-home operation, refer to p.45 and pages that follow.

Bit	Name	Setting	Initial value
0	Return-to-home operation selection	0: Use return-to-home operation selection type 1 1: Use return-to-home operation selection type 2	0
1 to 4	−	0: Fixed	−
5	Sensor mode selection*1	0: 2-sensor mode 1: 3-sensor mode	0
6	TIM signal used/not used *1	0: Not used 1: Used	0
7	External latch signal used/not used*1*2	0: Not used 1: Used	0
8 to 15	−	0: Fixed	−

*1 The set value becomes effective when the return-to-home operation selection bit is 1 (return-to-home operation type 2).
*2 The external latch signal selection conforms to the LT_SGNL setting specified by byte 2 of the ZRET command. If LT_SGNL=0 (TIM signal used), no external latch signal is used.

Setting example of return-to-home operation selection

Return-to-home operation type	Setting example
Type 1	00h
Type 2: 2-sensor mode, TIM signal not used	01h
Type 2: 2-sensor mode, TIM signal used	41h
Type 2: 3-sensor mode, TIM signal not used	21h
Type 2: 3-sensor mode, TIM signal used	61h

■ Resetting the absolute position loss error alarm

Reset the absolute position loss error alarm.
If an absolute position loss error alarm generates, change the setting of the absolute position loss error reset parameter from 0 to 1. The alarm will be reset and the coordinates will be confirmed.
The absolute position loss error alarm cannot be reset with the alarm/warning clear command (ALM_CLR: 06h).

■ Overtravel mask

Set whether or not ±LS signal detection alarms are output.
Whether the +LS input signal and −LS input signal are enabled or disabled is determined by the +LS mask and −LS mask settings, respectively, regardless of the overtravel mask setting.
Refer to p.56 for details.

9.6 Operation specifications

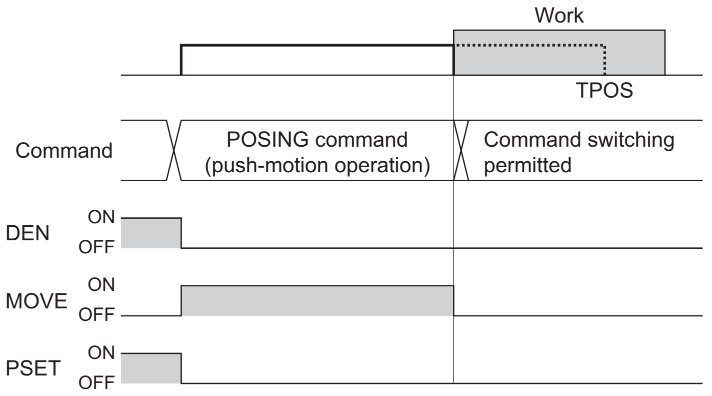
Timing charts of the push-motion operation using the POSING command and the return-to-mechanical-home operation type 2 are shown.

■ Push-motion operation using the POSING command

Note If you are not sure whether the option field can be set to the master, contact the manufacturer of your master device.

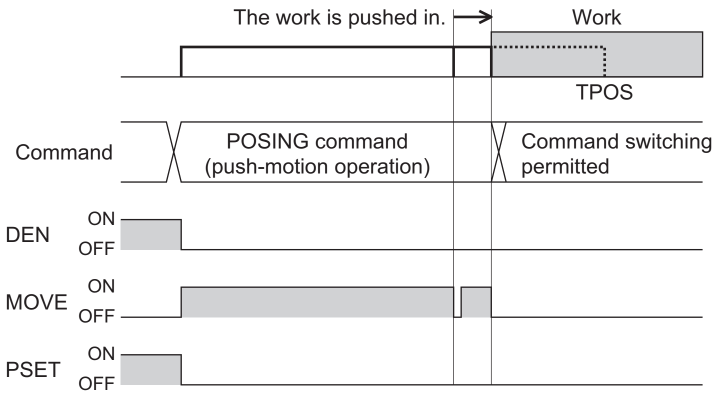
- The work is contacted before TPOS is reached while the POSING command is being executed

Since the DEN signal is still 0, completion of operation is determined by the MOVE signal.



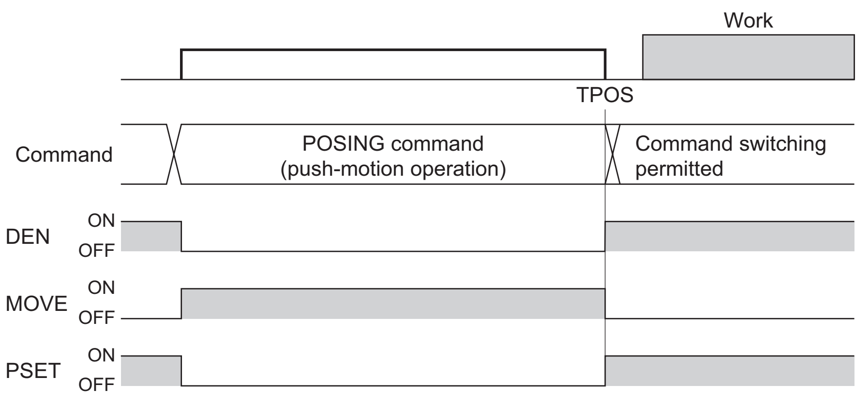
- The work is contacted and then pushed away slightly before TPOS is reached

The MOVE signal changes from 0 to 1 as the work is pushed away, and then changes back to 0 when the work is contacted again.



- Positioning operation is completed at the TPOS position without the work being contacted

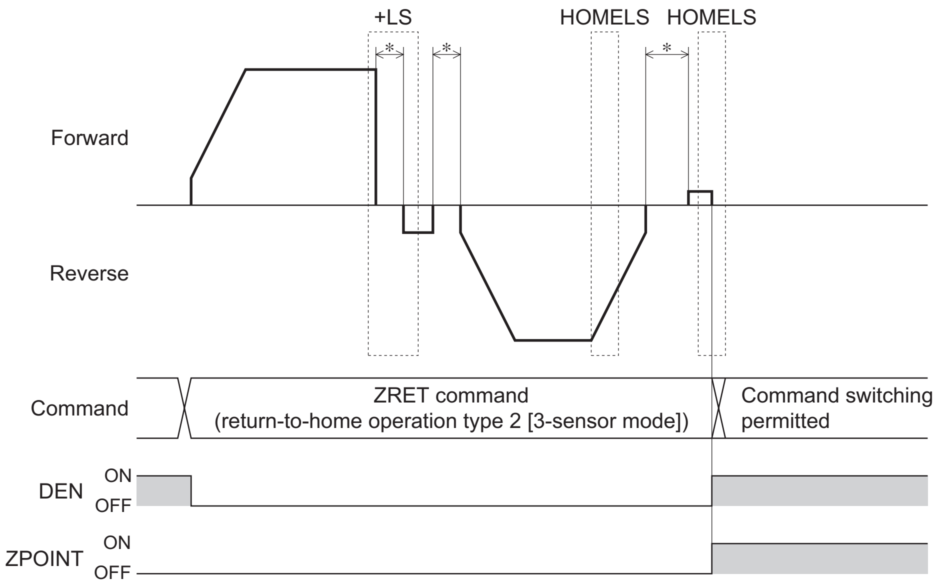
Completion of operation is determined by the DEN signal (change from 0 to 1).



■ Return-to-home operation using the ZRET command (return-to-home operation type 2)

- 3-sensor mode

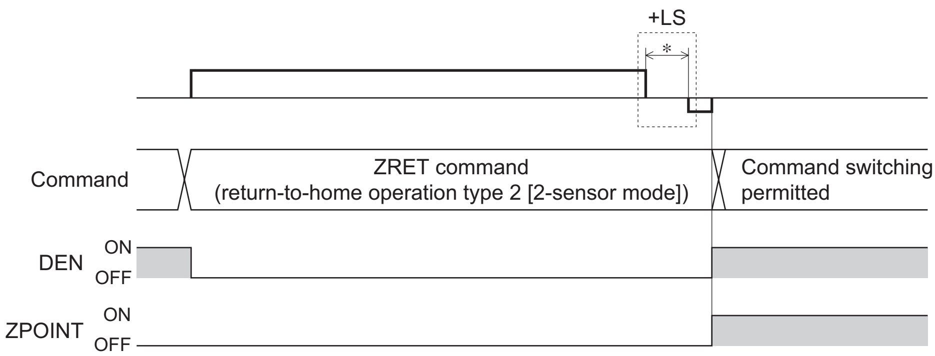
Completion of operation is determined by the DEN signal and ZPOINT signal



* When the sensor signal is detected, the motor will stop for 500 ms and then start moving to the opposite direction.

- 2 -sensor mode

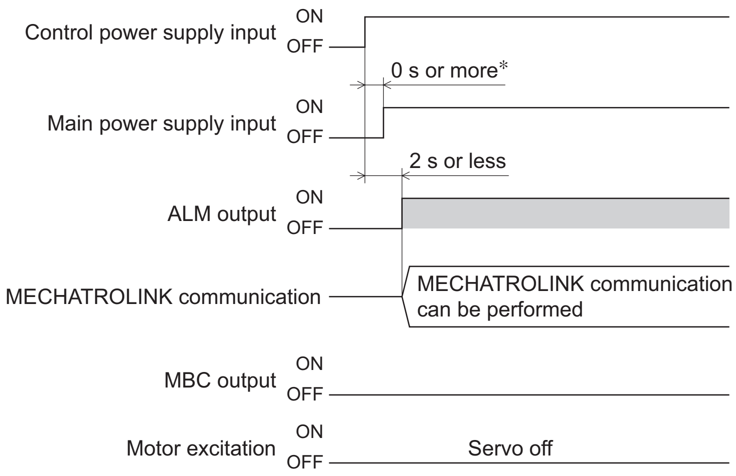
Completion of operation is determined by the DEN signal and ZPOINT signal



* When the sensor signal is detected, the motor will stop for 500 ms and then start moving to the opposite direction.

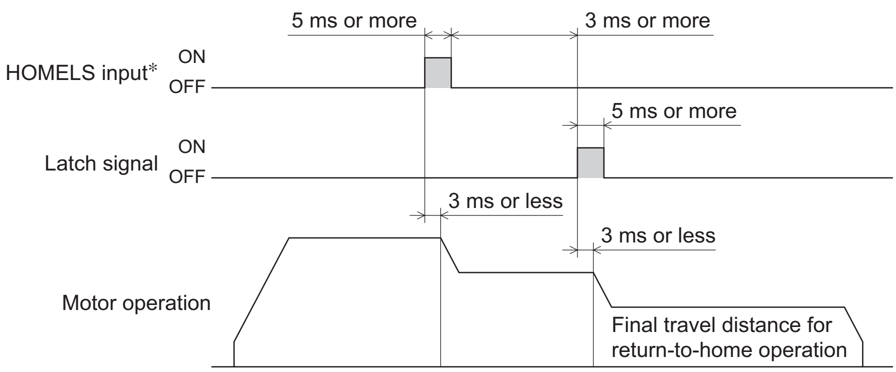
10 Timing charts

■ Power supply input



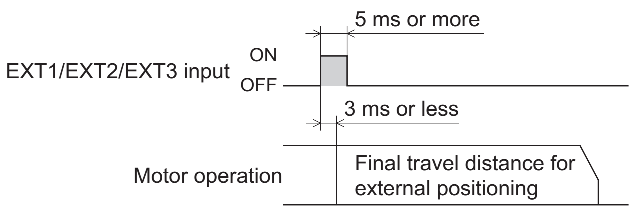
* Turn on the control power first.

■ Return-to-home operation

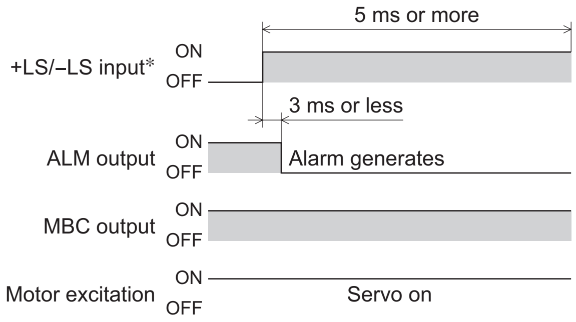


* Described based on active level.

■ EXT1/EXT2/EXT3 input (Positioning operation by external input)

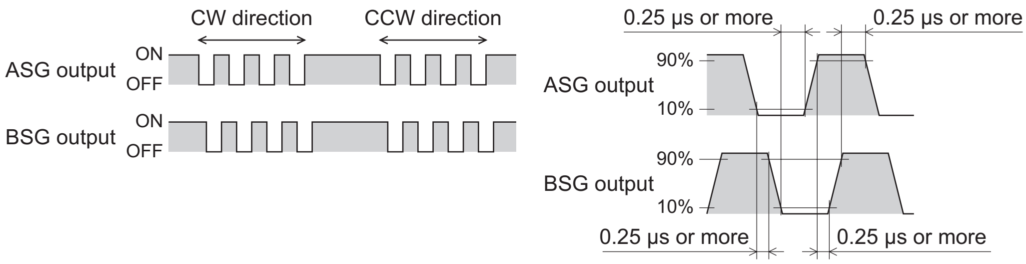


■ ±LS input



* Described based on active level.

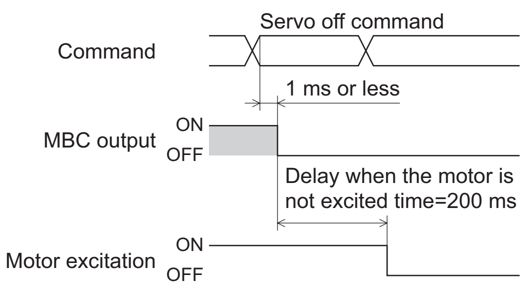
■ ASG/BSG output



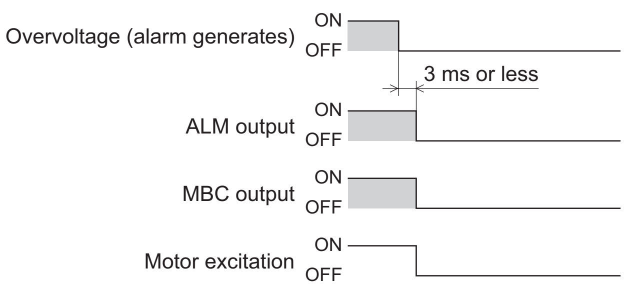
- The BSG output has a 90° phase difference with respect to the ASG output.
- The maximum output frequency is 500 kHz. (When the cable is 2 m [6.6 ft] long.)
- There is a maximum delay of 500 µs between ASG/BSG output and actual motor movement.
- The ASG/BSG output accuracy is within ±0.36° regardless of the resolution. (Repetitive positioning accuracy within ±0.09°)

■ MBC output

- When the servo off command is executed

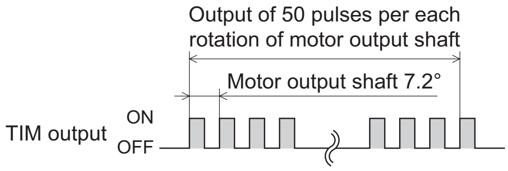


- When an alarm that cuts off motor current is present



■ TIM output

The TIM output turns ON every time the motor output shaft has turned 7.2°. (50 pulses are output per each rotation of the motor output shaft.)



Note

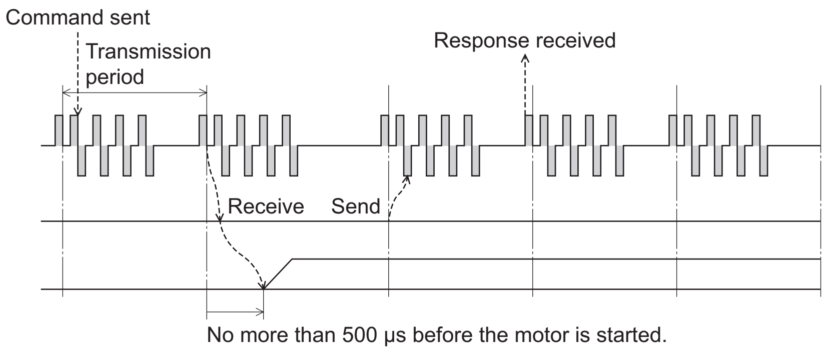
- If the TIM output is used as a latch signal, be sure to set the operation speed to 500 Hz or less.
- To switch the motor resolution using the electronic gear ratio, cycle the power or switch the resolution while the TIM output is ON and motor is stopped. If the motor resolution is changed while the TIM output is OFF or motor is operating, the TIM output may not turn ON even after the motor output shaft has turned 7.2°.

■ MECHATROLINK communication command/response timings

The following explains the timings at which MECHATROLINK communication commands are executed and monitored data are input. The same timings will apply regardless of the transmission period or communication period.

- Command data execution timing

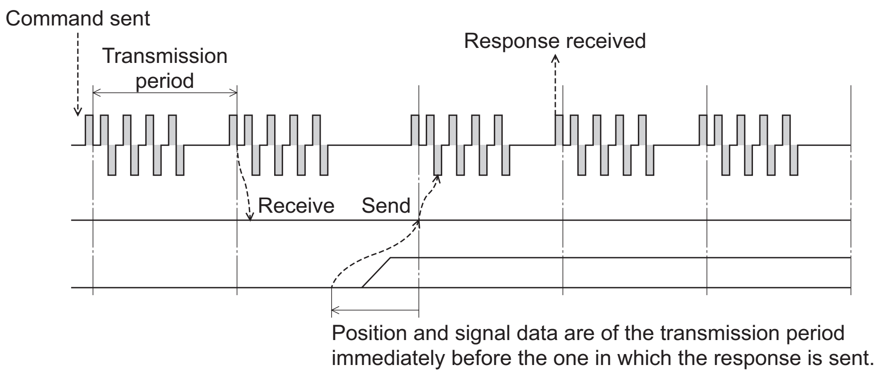
Motion commands and OPTION are executed within 500 µs upon command reception.



Command sent
Transmission period
Response received
No more than 500 ms before the motor is started

- Monitored data input timing

Monitor, I/O and status data are of the transmission period immediately before the one in which the response is sent.



11 Inspection

It is recommended that periodic inspections be conducted for the items listed below after each operation of the motor. If an abnormal condition is noted, discontinue any use and contact your nearest office.

■ During inspection

- Are any of motor mounting screws loose?
- Check for any unusual noises in the motor's bearings (ball bearings) or other moving parts.
- Are the motor's output shaft (or gear output shaft) and load shaft out of alignment?
- Are there any scratches, signs of stress or loose driver connection in the motor cable?
- Is there attachment of dust, etc., on the driver?
- Are any of the driver mounting screws or power connection terminal screws loose?
- Are there any strange smells or appearances within the driver?

Note The driver uses semiconductor elements, so be extremely careful when handling them. Static electricity may damage the driver.

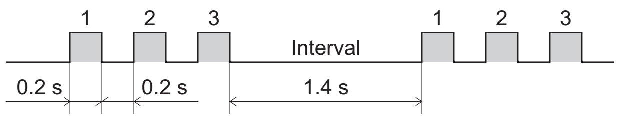
12 Troubleshooting and remedial actions

During motor operation, the motor or driver may fail to function properly due to an improper speed setting or wiring. When the motor cannot be operated correctly, refer to the contents provided in this section and take appropriate action. If the problem persists, contact your nearest office.

12.1 Driver alarm

When an alarm generates, the OP LED (red) will blink. At the same time, the ALM signal output will turn OFF and bit 0 (ALM) of the status field will change to 1.
If a warning is detected, the OP LED will remain a steady green light and the ALM signal output will not turn OFF. Only bit 1 (WARNG) of the status field will change to 1 and the operation will continue.

Example:
An alarm that cuts off the main power is detected (Number of OP LED blinks: 3)



■ Priority

If an alarm generates whose priority is higher than that of the present alarm, the current alarm code will be overwritten by the new alarm code of the higher priority. Alarms of the same or lower priority than the present alarm will be ignored. Alarms of priority 1 are not saved in the alarm history.

12.2 Limit switch pull-out sequence

If a ±LS detection error or soft limit detection error alarm generates, follow the procedure below to pull out from the limit:

1. Reset the alarm by sending the alarm/warning clear command (ALM_CLR: 06h) via MECHATROLINK communication with ALM_CLR_MOD=0.
   Use bit 0 (ALM) (=0) to check if the alarm has been cleared.
2. Pull out from the limit using one of the following methods:
   - Operate the motor in the direction in which it can be pulled out, using a motion command (INTERPOLATE, POSING, FEED, LATCH, EX_POSING or ZRET).
     Use bit 0 (+LS)/bit 1 (−LS) (=0) of the I/O monitor field or bit 12 (P_SOT)/bit 13 (N_SOT) (=0) of the status field to check if the motor has pulled out. If the motor was operated in the direction in which it cannot be pulled out, an alarm will generate again.
   - Turn off the motor excitation using the servo off command (SV_OFF: 32h) and move the motor in the direction in which it can be pulled out manually.
     If an electromagnetic brake is equipped, release the electromagnetic brake beforehand using the brake release request command (BRK_OFF).

12.3 Alarm code list

Number of OP LED blinks	COM LED	Alarm code	Phenomenon	Cause
Green solid	Red solid	01h	MECHATROLINK communication error	The executed command is not implemented.
Green solid	Red solid	02h	MECHATROLINK communication error	• The received command does not match the communication phase. • The execution condition of the command is not satisfied.
Green solid	Red solid	03h	MECHATROLINK communication error	Invalid command data due to one of the following reasons: · Outside the setting range · Outside the allowable range · Not supported
Red Two blinks	Green solid (phase 2, 3) or OFF	B1h	Overheat	The driver's heat-sink temperature reached approx. 85 °C (185 °F).
Red Two blinks	Green solid (phase 2, 3) or OFF	C0h	Overload	A cumulative load exceeding the maximum torque has been applied for the duration over the overload detection time.
Red Two blinks	Green solid (phase 2, 3) or OFF	C1h	Overspeed	The motor speed exceeded 5500 r/min.
Red 3 blinks	Green solid (phase 2, 3) or OFF	B2h	Overvoltage	The DC voltage of the main circuit is too high.
Red 3 blinks	Green solid (phase 2, 3) or OFF	B3h	Main power supply cut off detection	The main power was cut off.
Red 4 blinks	Green solid (phase 2, 3) or OFF	A0h	Excessive position deviation	The deviation between the command position and the actual position exceeded the set value of the overflow rotation amount of the motor output shaft.
Red 5 blinks	Green solid (phase 2, 3) or OFF	B0h	Overcurrent	The motor cable was shorted.
Red 6 blinks	Red solid	04h	MECHATROLINK communication error	Synchronous communication error The watchdog timer is not refreshed correctly.
Red 6 blinks	Red solid	05h	MECHATROLINK communication error	Transmission period setting error The transmission period set by the master is not supported.
Red 6 blinks	Red solid	06h	MECHATROLINK communication error	MECHATROLINK communication error A CRC error or other communication error generated at least twice consecutively.
Red 7 blinks	Green solid (phase 2, 3) or OFF	B7h	Insufficient ABS battery voltage (absolute specification only)	The ABS backup battery voltage dropped to, or below, the specified value.
Red 7 blinks	Green solid (phase 2, 3) or OFF	C3h	Absolute position loss (absolute specification only)	• Power was turned on for the first time after the battery was connected. • Battery is not connected or has been consumed. • The motor cable was disconnected when the main power was OFF. • The battery cable became open or a fuse was blown. • The multi-rotation operation range was exceeded.
Red 7 blinks	Green solid (phase 2, 3) or OFF	F0h	±LS signals logic error	Both the +LS and −LS sensors were detected in the ±LS signals enable mode.
Red 7 blinks	Green solid (phase 2, 3) or OFF	F1h	±LS signals reverse connection error	The LS signal on the side opposite to the operating direction was detected during a return-to-home operation in the 3-sensor mode or 2-sensor mode when the return-to-home operation selection bit was set to 1.
Red 7 blinks	Green solid (phase 2, 3) or OFF	F2h	Return-to-home operation error	The return-to-home sequence did not end properly.

How to reset the alarm
A: Cycle the main power and control power supply. (MECHATROLINK communication will cut off.)
B: Reset the alarm using the MECHATROLINK communication command's " ALM_CLR".
C: Reset the warning using the MECHATROLINK communication command's " ALM_CLR".

Remedial action	Motor action	Status field	Alarm reset	Priority
Check the command that has been sent.	Continue the operation. (The received command is ignored.)	Bit1: WARNG=1	C	1
Check the send condition of the command that has been sent.				
Check the data that has been sent.				
Check the ventilation condition inside the enclosure.	Cut off the motor current.	Bit0: ALM=1	B	3
Reduce the load or increase the acceleration/deceleration rate.				
Check the command speed or electronic gear.				
• Check the power supply voltage. • For a lifting device, reduce the load.				
Check to see if the main power is input correctly.				
Reduce the load or decrease the acceleration/deceleration rate.				
Check the motor cable and its connection to the driver.			A	4
Check if the watchdog timer is refreshed.			B	3
Set an appropriate transmission period within a range of 0.5 to 3.0 ms (in 0.5-ms increment).				
• Connect the MECHATROLINK communication cable correctly. • Connect a termination resistor correctly • Implement the noise elimination measures. • The driver may be damaged. Call our branch or sales office.				
Charge the battery. It takes approx. 48 hours to fully charge the battery at an ambient temperature of 20 °C (68 °F).	Stop the motor.			2
• Reset the alarm using the absolute position loss error reset parameter (1005h). • If the alarm still generates after charging the battery, a consumed battery, open cable or blown fuse may be the cause. Purchase a replacement battery (**PAEZ-BT2**).	Stop the motor.		D	2
Check the logic setting of the ±LS signals.	Stop the motor.		B	2
Check the ±LS signals wiring.				
• Check the installing/wiring the HOMELS signal and latch signal as well as return-to-home operation data. • An unanticipated load may have been applied during the return-to-home operation. Check the load.				

<table>
<tr><th>Number of OP LED blinks</th><th>COM LED</th><th>Alarm code</th><th>Phenomenon</th><th>Cause</th></tr>
<tr><td rowspan="5">Red
7 blinks</td><td rowspan="5">Green solid (phase 2, 3) or OFF</td><td>F3h</td><td>HOMELS signal non-detection error</td><td>The HOMELS signal was not detected at a position between +LS signal and −LS signal.</td></tr>
<tr><td>F4h</td><td>Latch signal non-detection error</td><td>The latch signal was not detected at a position between +LS signal and −LS signal.</td></tr>
<tr><td>F6h</td><td>±LS signals detection error</td><td>The +LS or −LS signal was detected in the ±LS signal-enable mode.</td></tr>
<tr><td>F7h</td><td>Soft limit detection error</td><td>The motor reached a soft limit position.</td></tr>
<tr><td>FAh</td><td>Return-to-home final travel operation error</td><td>The ±LS signal was detected during operation over the final travel distance for return-to-home operation.</td></tr>
<tr><td rowspan="3">Red
8 blinks</td><td rowspan="7">Green solid (phase 2, 3) or OFF</td><td>B8h</td><td>Sensor error</td><td>A sensor error was detected during operation.</td></tr>
<tr><td>D2h</td><td>Sensor error</td><td>A sensor error occurred when the power was turned on (motor cable not connected, etc.).</td></tr>
<tr><td>D3h</td><td>Rotor rotation at initialization</td><td>Initialization failed because the motor was rotating when the power was turned on.</td></tr>
<tr><td rowspan="4">Red
9 blinks</td><td>82h</td><td>MECHATROLINK communication ASIC error</td><td>An error occurred with the MECHATROLINK communication ASIC.</td></tr>
<tr><td>B9h</td><td>ABS backup system error</td><td>An error was occurred with ABS backup system.</td></tr>
<tr><td rowspan="2">D1h</td><td rowspan="2">Nonvolatile memory error</td><td>Stored data for the driver was damaged.</td></tr>
<tr><td>Rewrite life of the NV memory (approx. 100,000 times) was reached.</td></tr>
</table>

How to reset the alarm
A: Cycle the main power and control power supply. (MECHATROLINK communication will cut off.)
B: Reset the alarm using the MECHATROLINK communication command's " ALM_CLR".
C: Reset the warning using the MECHATROLINK communication command's " ALM_CLR".

Remedial action	Motor action	Status field	Alarm reset	Priority
• Set a HOMELS signal between the +LS signal and −LS signal. • Check the starting direction of return-to-home operation.	Stop the motor.	Bit0: ALM=1	B	2
• Position the sensors so that the latch signal will be detected after the HOMELS signal is detected. • Check the starting direction of return-to-home operation.				
After resetting the alarm, pull out from the sensor position by referring to “12.2 Limit switch pull-out sequence” on p.67.				
Check the command position exceeds the soft limits.				
• Check the final travel distance for return-to-home operation. • Check the ±LS signals installation position.				
Turn off the power and check the motor cable and its connection to the driver	Cut off the motor current.		A	4
Turn off the power and check the motor cable and its connection to the driver				
Prevent the motor output shaft from rotating due to an external force when the power is turned on.				
The communication IC may be damaged. Call our branch or sales office.				
Cycle the power.				
Initialize the parameter.				
The NV memory may be damaged. Call our branch or sales office.				

13 Specifications

■ General specifications

		Motor	Driver
Protective range		IP20	IP10
Operation environment	Ambient temperature	−10 to +50 °C (+32 to +122 °F) (non-freezing) Harmonic geared type: 0 to +40 °C (+32 to +104 °F) (non-freezing)	0 to +40 °C (+32 to +104 °F) (non-freezing)
	Humidity	85% or less (non-condensing)	
	Altitude	Up to 1000 m (3300 ft.) above sea level	
	Surrounding atmosphere	No corrosive gas, dust, water or oil	
Storage environment	Ambient temperature	−20 to +60 °C (−4 to +140 °F) (non-freezing)	−25 to +70 °C (−13 to +158 °F) (non-freezing)
	Humidity	85% or less (non-condensing)	
	Altitude	Up to 3000 m (10000 ft.) above sea level	
	Surrounding atmosphere	No corrosive gas, dust, water or oil	
Shipping environment	Ambient temperature	−20 to +60 °C (−4 to +140 °F) (non-freezing)	−25 to +70 °C (−13 to +158 °F) (non-freezing)
	Humidity	85% or less (non-condensing)	
	Altitude	Up to 3000 m (10000 ft.) above sea level	
	Surrounding atmosphere	No corrosive gas, dust, water or oil	

■ Battery

Battery type	Cylindrical sealed nickel/cadmium rechargeable battery
Nominal voltage	2.4 V
Rated capacity	2000 mAh
Weight	0.18 kg (0.396 lb.)
Life	Approx. 4 years*1
Charge time	48 hours*1
Data retention period*1*2	15 days
Ambient temperature	0 to +40 °C (+32 to +104 °F) (non-freezing)
Ambient humidity	20 to 85%RH (non-condensing)
Protective circuit	Fuse
Storage temperature	−20 to +45 °C (−4 to +113 °F) (under 3 months) −20 to +35 °C (−4 to +95 °F) (3 months or more)

*1 At an ambient temperature of 20 °C (68 °F)
*2 After the power is cut off with the battery fully charged.

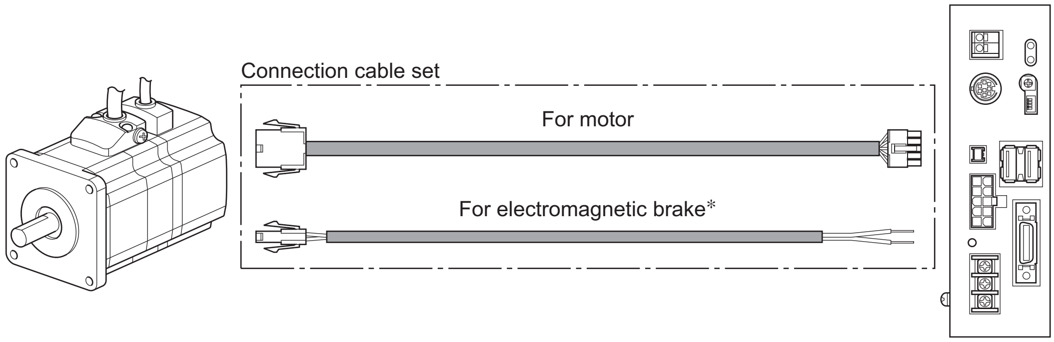
14 Options (sold separately)

■ Motor connection cable

The cable supplied with the **ARL** series is all you need to connect the motor and driver.
Take note, however, that if you wish to connect the motor and driver over a distance of 3 m (9.8 ft.) or more, the supplied cable is not long enough and you must use a connection cable set or extension cable set.
When installing the motor on a moving part, use a flexible cable offering excellent flexibility.

- Extending the wiring length using a connection cable set

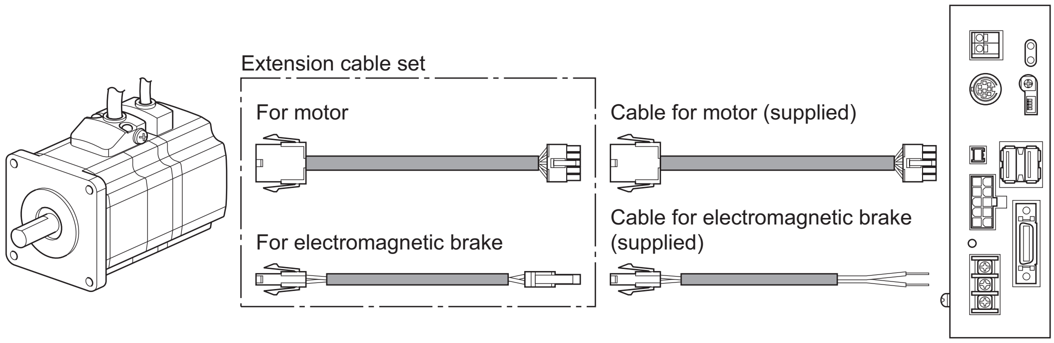
Do not use the supplied cable.



* Only when the motor is of electromagnetic brake type.

- Extending the wiring length using an extension cable set

These can be used with electromagnetic brake motors. Connect an extension cable to the supplied cable.
With standard motors, extend the wiring using the connection cable set.



Note When extending the wiring length by connecting an extension cable to the supplied cable, keep the total cable length to 20 m (65.6 ft.) or less.

- Cable set types

Motor type	Connection cable set	Extension cable set
For standard motor	For motor	Not available. Use motor connection cable.
For electromagnetic brake motor	• For motor • For electromagnetic brake	• For motor • For electromagnetic brake

- Connection cable set

A cable is needed to connect the motor and driver.
The cable set for electromagnetic brake motor consists of two cables, one for motor and the other for electromagnetic brake.

- For standard motor

Model	Length [m (ft.)]
CC010VA2F	1 (3.3)
CC020VA2F	2 (6.6)
CC030VA2F	3 (9.8)
CC050VA2F	5 (16.4)
CC070VA2F	7 (23)
CC100VA2F	10 (32.8)
CC150VA2F	15 (49.2)
CC200VA2F	20 (65.6)

- For electromagnetic brake motor

Model	Length [m (ft.)]
CC010VA2FB	1 (3.3)
CC020VA2FB	2 (6.6)
CC030VA2FB	3 (9.8)
CC050VA2FB	5 (16.4)
CC070VA2FB	7 (23)
CC100VA2FB	10 (32.8)
CC150VA2FB	15 (49.2)
CC200VA2FB	20 (65.6)

- Flexible connection cable set

A cable offering excellent flexibility, necessary for connecting the motor and driver.
The flexible cable set for electromagnetic brake motor consists of two cables, one for motor and the other for electromagnetic brake.

- For standard motor

Model	Length [m (ft.)]
CC010VA2R	1 (3.3)
CC020VA2R	2 (6.6)
CC030VA2R	3 (9.8)
CC050VA2R	5 (16.4)
CC070VA2R	7 (23)
CC100VA2R	10 (32.8)
CC150VA2R	15 (49.2)
CC200VA2R	20 (65.6)

- For electromagnetic brake motor

Model	Length [m (ft.)]
CC010VA2RB	1 (3.3)
CC020VA2RB	2 (6.6)
CC030VA2RB	3 (9.8)
CC050VA2RB	5 (16.4)
CC070VA2RB	7 (23)
CC100VA2RB	10 (32.8)
CC150VA2RB	15 (49.2)
CC200VA2RB	20 (65.6)

- Extension cable set

This cable is needed to extend the wiring length between the motor and driver.
The extension cable set consists of two cables, one for motor and the other for electromagnetic brake.

- For electromagnetic brake motor

Model	Length [m (ft.)]
CC010VA2FBT	1 (3.3)
CC020VA2FBT	2 (6.6)
CC030VA2FBT	3 (9.8)
CC050VA2FBT	5 (16.4)
CC070VA2FBT	7 (23)
CC100VA2FBT	10 (32.8)
CC150VA2FBT	15 (49.2)

- Flexible extension cable set

This cable offering excellent flexibility is needed to extend the wiring length between the motor and driver.
The flexible extension cable set consists of two cables, one for motor and the other for electromagnetic brake.

- For electromagnetic brake motor

Model	Length [m (ft.)]
CC010VA2RBT	1 (3.3)
CC020VA2RBT	2 (6.6)
CC030VA2RBT	3 (9.8)
CC050VA2RBT	5 (16.4)
CC070VA2RBT	7 (23)
CC100VA2RBT	10 (32.8)
CC150VA2RBT	15 (49.2)

■ Data setting software

The data setting software lets you set operation data and parameters using a PC.
The software comes with a PC interface cable [5 m (16.4 ft.)]. The cable is connected to the USB port on the PC.

Model: **MEXE02**

■ Battery set

A set of a battery and a battery holder used in the absolute specification. A battery and a battery holder can be purchased separately.

Model: **PAEZ-BT2H** (battery and battery holder)
PAEZ-BT2 (battery)
PAEZ-BTH2 (battery holder)

■ DIN rail mounting plate

Plate for mounting the driver to a DIN rail [35 mm (1.38 in.)].

Model: **PADP01**

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• Please contact your nearest Oriental Motor office for further information.

ORIENTAL MOTOR U.S.A. CORP.
Technical Support Tel:(800)468-3982
8:30 A.M. to 5:00 P.M., P.S.T. (M-F)
7:30 A.M. to 5:00 P.M., C.S.T. (M-F)
www.orientalmotor.com

ORIENTAL MOTOR DO BRASIL LTDA.
Tel:+55-11-3266-6018
www.orientalmotor.com.br

ORIENTAL MOTOR (EUROPA) GmbH
Schiessstraße 44, 40549 Düsseldorf, Germany
Technical Support Tel:00 800/22 55 66 22
www.orientalmotor.de

ORIENTAL MOTOR (UK) LTD.
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Tel:080-777-2042
www.inaom.co.kr

ORIENTAL MOTOR CO., LTD.
Hong Kong Branch
Tel:+852-2427-9800

ORIENTAL MOTOR CO., LTD.
4-8-1 Higashiueno, Taito-ku, Tokyo
110-8536 Japan
Tel:03-6744-0361
www.orientalmotor.co.jp